Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
JPLPC – Malvar Campus
G. Leviste St. Poblacion, Malvar, Batangas
Tel. Nos.: (043) 778-2170/ (043) 406-0830 loc. 9204
E-mail Address: cics.malvar@g.batstate-u.edu.ph | Website Address: http://www.batstate-u.edu.ph

COLLEGE OF INFORMATICS AND COMPUTING SCIENCES

AEGIS: An AI – Enhanced Guidance and Integrated School Information System for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas Batangas Inc.

A Capstone Project Presented to the Faculty of the
College of Informatics and Computing Sciences
Batangas State University
The National Engineering University
JPLPC – Malvar Campus
Malvar, Batangas

In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology
with Specialization in Business Analytics

NARVACAN, JHAREN B.
VILLANUEVA, STEPHANIE LOREINE M.
VILLEGAS, JUSTINE LAY J.

NOEL L. VIRREY, BSCS
Adviser

MAY 2026

Republic of the Philippines
BATANGAS STATE UNIVERSITY
The National Engineering University
JPLPC – Malvar Campus
COLLEGE OF INFORMATICS AND COMPUTING SCIENCES

APPROVAL SHEET

This Capstone Project entitled **"AEGIS: An AI – Enhanced Guidance and Integrated School Information System for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas Batangas Inc."** prepared and submitted by **NARVACAN, JHAREN B., VILLANUEVA, STEPHANIE LOREINE M.,** and **VILLEGAS, JUSTINE LAY J.,** in partial fulfillment of the requirements for the Degree, **Bachelor of Science in Information Technology with Specialization in Business Analytics,** has been examined and is recommended for Oral Examination.

NOEL L. VIRREY, BSCS
Adviser

Approved by the Committee on Oral Examination with a grade of _____.

PANEL OF EXAMINERS

SHIELA MARIE G. GARCIA, MSCS
Chairperson

CRUZZETE A. CALZO, BSCS
Member

EDDIE D. BUCAD, BSIT
Member

Accepted and approved in partial fulfillment of the requirements for the Degree, **Bachelor of Science in Information Technology with Specialization in Business Analytics.**

_____
Date

SHIELA MARIE G. GARCIA, MSCS
College Dean

TABLE OF CONTENTS

TITLE **PAGE**

TITLE PAGE……...i

TABLE OF CONTENTS……x

LIST OF FIGURES……....xv

LIST OF TABLES……..xix

INTRODUCTION 1

Project Context 1

Purpose and Description 4

Objectives of the Study 8

Scope and Limitations of the Study .....**Error! Bookmark not defined.**

Definition of Terms 14

REVIEW OF RELATED SYSTEMS..... 18

Technical Background..... 18

System Architecture 25

Related Systems 30

DESIGN AND METHODOLOGY..... 45

Data Gathering 45

Project Concept 45

Development Method 46

Development Approach 55

System Analysis and Design 57

Context Flow Diagram 57

Data Flow Diagram 59

Functional Requirements 64

Non-Functional Requirements 74

User Design Interface 77

Hardware Requirements in Developing the System 106

Hardware Requirements in Using the System 107

Software Requirements in Developing the System 109

Software Requirements in Using the System 106

Testing and Evaluation 106

Method of Testing 118

BIBLIOGRAPHY 125

LIST OF FIGURES

FIGURE		PAGE
1	System Architecture Diagram of AEGIS	32
2	Agile Methodology	52
3	Top-Down Approach	57
4	Context Flow Diagram of the AEGIS	60
5	Data Flow Diagram of the Admin of AEGIES	62
6	Data Flow Diagram of The Faculty of AEGIS	64
7	Data Flow Diagram of The Students of AEGIS	71
8	Data Flow Diagram of The Parents of AEGIS	73
9	Data Flow Diagram of The Guidance of AEGI	74
10	Login Interface	75
11	Home Dashboard	76
12	Academic Assessment Interface	77
13	Analytics Interface	78
14	Hamburger Menu Interface	80
15	Profile Interface	81

16 AI Chatbot Interface 83

17 Settings Interface 84

18 Parents Dashboard Interface 85

19 Student Information Interface 87

20 Parent Profile Interface 88

21 Web Login Interface 89

22 Faculty Web Dashboard Interface 91

23 Student Profile Interface 92

24 Attendance Monitoring Interface 93

25 Behavior Notes Interface 94

26 Subject & Scheduling Interface 96

27 Grade Encoding Interface 97

28 Guidance Counselor Web Dashboard 98

29 Guidance Counselor Student Profile Interface 100

30 Consultation Interface 101

31 Assessment Interface 103

32 Role and Access Interface 104

33 Personal Information Interface 105

34 Project Timeline of AEGIS 127

LIST OF TABLES

TABLE **PAGE**

1 Functional Requirements .....
2 Non-Functional Requirements 68
3 Hardware Requirements for the Developers 107
4 Hardware Requirements For The Users 108
5 Hardware Requirements For The Users 109
6 Software Requirements For The Developers..... 110
7 Software Requirements For The Users..... 111
8 Likert Scale..... 113
9 Scale Range and Verbal Interpretation 114
10 Sample Test Case Result for Student..... 115
11 Sample Test Case Result for Administrators 119

CHAPTER 1

INTRODUCTION

This chapter discusses the rationale behind the proposed system. It covers the project context, which describes the current situation and existing gaps in student monitoring. The chapter also presents the purpose and description of the system, the objectives of the study, its scope and limitations, and the definition of key terms used in the research.

Project Context

Schools today are expected to do more than simply deliver instruction. They are responsible for maintaining accurate records of student performance, keeping parents informed of their child's academic progress, and identifying students who may need additional support before their difficulties become severe. Meeting these responsibilities effectively requires systems that are organized, accessible, and responsive to the needs of everyone in the school community. According to Mfanafuthi et al., the availability of centralized school information systems is directly associated with improvements in administrative efficiency, stakeholder engagement, and the overall quality of educational services delivered to students and their families [1]. In the Philippines, however, many private high schools have yet to build this kind of infrastructure, and continue to rely on manual and

fragmented processes that make it difficult to fulfill these responsibilities consistently and accurately.

HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas Inc. is one such institution. A preliminary interview conducted with the school's faculty adaministrator revealed that academic records at the institution are managed through separate tools and physical files, with no unified system in place to consolidate them. Grades, attendance, and behavioral notes are maintained independently of one another, requiring teachers and class advisers to manually gather and reconcile information from multiple sources whenever an official report needs to be produced. This process alone demands approximately one to two days of work from teachers before every card-giving day, time that is spent on administrative tasks rather than on instruction and direct student engagement.

These gaps in record management also affect how information reaches the people who need it most. Parents of enrolled students are typically informed of academic or behavioral concerns only when a class adviser personally contacts them or when a school-wide announcement is issued through the guidance office. This arrangement limits parental awareness to irregular, reactive updates rather than consistent and accessible information. Studies have shown that sustained parental involvement in a child's education is associated with improved academic performance and overall student well-being [2], yet such involvement is difficult to

maintain when parents have no direct and reliable means of monitoring their child's progress. Students face a similar limitation, as accessing personal academic records currently requires a physical visit to the school, with no self-service portal available for their use. The challenges extend further into the guidance office, where consultation records, session logs, and behavioral incident documentation are currently maintained without the support of a dedicated digital system. The absence of structured documentation makes it difficult to track the history of student concerns over time, monitor the progress of ongoing interventions, and generate reports needed for follow-up and institutional planning. These limitations reduce the guidance counselor's capacity to respond comprehensively to the needs of students requiring support.

Perhaps the most consequential gap in the institution's current system involves the identification of students who are struggling academically. At present, this process depends on the direct observation of class advisers and subject teachers, who are responsible for reporting at-risk students to the school head so that they may be referred to their existing intervention program like remedial program. While this reflects the institution's genuine commitment to student support, research has consistently shown that academic interventions are significantly more effective when delivered early, before a student's difficulties have deepened [3]. Without a mechanism that systematically analyzes student performance data, the identification

of at-risk students remains reactive rather than anticipatory, and struggling learners may go undetected until the point when timely intervention is no longer possible.

The integration of digital school management systems supported by artificial intelligence has been demonstrated to address precisely these kinds of challenges. According to Tugano et al., centralized platforms reduce the duplication of administrative work, improve the accuracy and accessibility of student records, and provide stakeholders with timely information through role-based portals [4]. AI-driven tools further extend what these systems can offer, enabling schools to move beyond manual observation and toward data-informed approaches to identifying students who may need intervention and supporting the decisions of guidance counselors and administrators [5].

In response to these documented challenges, the proponents propose the development of AEGIS: AI-Enhanced Guidance and Integrated School Information System for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas Inc. AEGIS is a web-based platform designed to consolidate academic records, attendance, schedules, announcements, and guidance services within a single organized environment accessible to high school students, parents, faculty, guidance counselors, and administrators. The system incorporates artificial intelligence capabilities aimed at reducing the administrative burden on teachers, providing stakeholders with timely and direct access to school information, and

enabling a proactive, data-driven approach to identifying and supporting students who may be at risk of academic difficulty. Through these features, AEGIS seeks to address the operational and communicative gaps currently experienced by HIS CARE CHRISTIAN INSTITUTE and to support the institution in fulfilling its responsibility to every student it serves.

Purpose and Description

This project aims to improve the academic and administrative management practices of HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas Inc. by introducing AEGIS, an AI-Enhanced Guidance and Integrated School Information System developed specifically for the institution's high school community. The system delivers a comprehensive web-based solution built around the operational needs of students, parents, faculty, guidance counselors, and administrators, with the goal of centralizing school information, reducing administrative workload, and enabling a more informed and responsive approach to student support.

The main purpose of AEGIS is to provide a unified platform that brings together the school's academic, administrative, and guidance functions within a single organized and secure environment. Through the system, teachers and class advisers are able to manage student profiles, record and monitor attendance, encode grades, and generate reports without relying on separate tools or physical

documents. A batch file processing feature further supports efficient data management by allowing records to be uploaded and consolidated from the school's existing files, reducing the time teachers spend on administrative preparation and freeing them to focus more directly on instruction.

Beyond record management, AEGIS provides dedicated portals for each type of user in the school community. Students are able to access their own academic records, view attendance and behavioral notes, receive school announcements, and interact with the system's AI Chatbot for academic assistance without needing to visit the school physically. Parents, in turn, are given direct and continuous visibility into their child's academic performance, attendance, and behavioral concerns through a portal that keeps them consistently informed rather than dependent on irregular updates from class advisers. These portals address the communication gaps identified in the institution's current processes by making relevant information accessible to the people who need it most, at any time and through a browser.

The Guidance Module of AEGIS is designed to support the guidance counselor in managing student concerns more systematically. The module enables the documentation of consultation records, session logs, and behavioral incident reports within the platform, providing a structured record of each student's history that can be referenced for follow-up and reporting. Guidance counselors are also able to create and deploy student assessments through the system, with the AI

Chatbot available to assist in generating assessment questions based on the student's profile and needs. These features give the guidance office a more organized and efficient foundation for delivering student support services.

One of the most significant capabilities of AEGIS lies in its analytical and AI-powered functions. Abdul Razak et al., demonstrated the feasibility of deploying NLP-based conversational agents in educational settings, developing a college assistance chatbot that achieved an accuracy of 80.07% in processing and responding to a wide range of student queries [6]. Building on this foundation, AEGIS integrates an AI Chatbot powered by Natural Language Processing that enables intelligent query handling across all user roles, understanding and responding to user concerns, routing escalated issues to the appropriate modules or personnel, and recommending relevant academic resources based on student data. The system further extends this capability by embedding the chatbot within a full school information management platform, allowing it to draw from consolidated academic and behavioral records in generating its responses.

Muraina et al., similarly established that Decision Tree algorithms are effective in identifying students at risk of poor academic performance by analyzing patterns in course data, providing a data-driven basis for early intervention [7]. AEGIS builds on this finding by employing a Decision Tree algorithm that analyzes patterns across all high school subjects using imported academic and behavioral

data, generating proactive alerts for guidance counselors before a student's difficulties become severe. Collaborative Filtering is also integrated to produce personalized academic recommendations tailored to each student's performance profile, supporting more targeted and meaningful intervention. Together, these AI capabilities allow AEGIS to move beyond passive data storage and toward a system that actively supports decision-making at every level of the school community.

The Administrator Module provides school administrators with tools for managing user accounts across all roles, controlling system-wide configurations, and overseeing the school's content and announcements through a built-in Content Management System. Data import management is also available, allowing administrators to upload and process CSV or Excel files exported from the school's existing records and map them to the appropriate system entries for students, grades, and attendance. This capability ensures that AEGIS can be adopted without requiring the institution to abandon its existing data infrastructure.

To protect the sensitive information of students and all other users, AEGIS operates under a security framework that includes role-based access control, password encryption, and audit logs. All data handled within the system is managed in full compliance with the Data Privacy Act of 2012 (Republic Act No. 10173), ensuring that the privacy and rights of every stakeholder are upheld throughout the system's operation. Through these features, AEGIS seeks not only to modernize

how HIS CARE CHRISTIAN INSTITUTE manages its school information, but to create a more connected, proactive, and student-centered environment where technology serves the genuine needs of everyone it is built to support.

Objectives of the Study

The general objective of this study is to design, develop, and evaluate AEGIS, a web-based school information and management system with AI-powered features for the high school students, parents, faculty, guidance counselors, and administrators of HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas Inc., aimed at streamlining academic processes and enhancing data-driven decision-making in a secure and data privacy-compliant environment.

The following are the specific objectives of the study.

1. To develop a Faculty and Advisory Class Module that enables teachers and advisers to:

    1.1 Manage advisory class records including:

        1.1.1 Student profiles and basic information:

        1.1.2 Attendance monitoring and recording;

        1.1.3 Behavioral notes and class-related concerns;

    1.2 Manage academic records including:

        1.2.1 Subject assignments and class scheduling;

        1.2.2 Grade encoding and class record management;

1.2.3 Batch file processing for efficient uploading of grades and records.

1.3 Generate subject performance reports, class summaries, and advisory reports for submission.

2. To develop a Student Module that enables high school students to:

2.1 Access personal academic information including:

2.1.1 Student profiles and basic information;

2.1.2 Attendance monitoring and recording;

2.1.3 Behavioral notes and class-related concerns.

2.2 Receive school-related updates including:

2.2.1 Announcements and school news;

2.2.2 Real-time notifications on grades, attendance, and concerns.

2.3 Interact with the AI Chatbot for academic assistance and personalized recommendations;

2.4 Take AI-generated assessments provided by the guidance counselor through the system.

3. To develop a Parent/Guardian Module that allows parents to:

3.1 Monitor their child's academic information including:

3.1.1 Real-time academic performance and grades;

3.1.2 Attendance records and behavioral concerns.

4. To develop a Guidance Module that supports guidance counselors in:

4.1 Managing student consultation records including:

4.1.1 Session logs and consultation history;

4.1.2 Behavioral incident records and intervention tracking;

4.2 Generating and administering student assessments using the system, including:

4.2.1 Creating assessment questions with the assistance of the AI Chatbot;

4.2.2 Deploying assessments to students for completion within the platform;

4.2.3 Monitoring student assessment results for tracking and follow-up

4.3 Receiving AI-generated alerts for at-risk students based on:

4.3.1 Academic performance patterns detected through a Decision Tree Algorithm;

4.3.2 Query escalation flags raised from AI Chatbot interactions using NLP-based topic classification.

4.4 Generating individual student guidance reports for documentation and follow-up;

5. To develop an Administrator Module equipped with:

5.1 A Content Management System (CMS) for managing:

5.1.1 School announcements, news, and resources;

5.1.2 System-wide configurations and module controls.

5.2 User account management for all system roles including students, parents, faculty, guidance counselors, and administrators;

5.3 Data import management including:

5.3.1 Uploading and processing CSV/Excel files exported from the school; and

5.3.2 Mapping imported data to system records for students, grades, and attendance.

6. To develop an Analytical Function that processes imported data and utilizes an integrated AI Chatbot that will:

6.1 Provide visual reports and graphs on school-wide academic performance and attendance trends based on imported data;

6.2 Generate AI-driven recommendations using Collaborative Filtering and Decision Tree Algorithm to support data-driven decision-making;

6.3 Enable an AI Chatbot that will:

6.3.1 Process and understand user queries using Natural Language Processing (NLP) across all user roles;

6.3.2 Analyze query nature and urgency using NLP-based topic classification to route concerns to appropriate modules or personnel;

6.3.3 Suggest academic resources and learning materials based on imported student performance data;

6.3.4 Recommend intervention strategies for guidance counselors based on imported student behavioral and academic patterns;

6.3.5 Assist guidance counselors in generating assessment questions for student evaluation;

6.3.6 Provide personalized responses based on user role and historical interaction data;

6.4 Produce exportable reports for official documentation and reporting purposes.

7. To test and evaluate the system based on ISO/IEC 25010:2011 software quality standards by:

7.1 Functionality;

7.2 Reliability;

7.3 Usability;

7.4 Efficiency;

7.5 Maintainability;

7.6 Data Privacy; and

7.7 Security.

Scope and Limitations of the Study

This study aims to design, develop, and evaluate AEGIS, an AI-Enhanced Guidance and Integrated School Information System developed as a web-based platform for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas Inc. The development of the system is conducted within the duration of the academic project period. The primary users of the system are high school students, parents, faculty, guidance counselors, and administrators, each provided with a dedicated role-based portal that grants access only to the functions and information relevant to their respective responsibilities within the school.

The scope of the system includes academic record management covering grade encoding, attendance monitoring, and behavioral note documentation, as well as class scheduling, subject assignments, and batch file processing for the efficient uploading of student records through CSV or Excel files exported from the school's existing records. Faculty and class advisers are able to manage advisory class records and generate subject performance reports and class summaries through the platform. Students are provided with a self-service portal for accessing their personal academic information, receiving real-time notifications, and interacting with the AI Chatbot for academic assistance. Parents are given visibility into their child's academic performance, attendance, and behavioral concerns through a dedicated monitoring portal. The Guidance Module supports the documentation of

consultation records, session logs, and behavioral incident reports, as well as the creation and deployment of student assessments with AI-assisted question generation.

The system also covers an analytical function that includes visual reports and graphs on school-wide academic performance and attendance trends. AI-powered features are incorporated in two forms: features that operate on imported data, including a Decision Tree algorithm for proactive at-risk student identification and Collaborative Filtering for generating personalized academic recommendations; and a real-time AI Chatbot powered by Natural Language Processing that processes user queries, routes escalated concerns to appropriate modules or personnel, and recommends academic resources based on the school content managed through the system's Content Management System. The Administrator Module covers user account management, content management for school announcements and resources, and data import management for CSV and Excel files. All system functions operate under a security framework compliant with the Data Privacy Act of 2012 (Republic Act No. 10173), including role-based access control, password encryption, and audit logs. The system is also designed to accommodate all enrolled students and registered users of the institution simultaneously, ensuring that all stakeholders can access the platform without restrictions on the number of concurrent users.

AEGIS is specifically developed for the high school department of HIS CARE CHRISTIAN INSTITUTE and is not intended to cover the operations of other educational levels or institutions without further customization. The system does not include features for financial management, tuition processing, or library management, as the scope of the project is limited to academic records, guidance services, and AI-powered student support. As a web-based platform, AEGIS requires a stable internet connection at all times and does not support offline access. Users without internet connectivity will be unable to access any function of the system, and performance may be further affected by network instability or insufficient device specifications on the part of the user.

The accuracy and effectiveness of the Decision Tree and Collaborative Filtering features are dependent on the completeness and correctness of the data imported into the system. Incomplete, outdated, or inaccurate records may affect the reliability of the at-risk alerts and academic recommendations generated by these features. The AI Chatbot's responses are likewise limited to the scope of the academic content and resources managed through the system's Content Management System, and its performance may be limited in handling concerns that fall outside the school's available data. The data import function of AEGIS supports CSV and Excel file formats only and does not integrate directly with any external LMS or third-party platform.

Furthermore, the system is developed within the specific context of HIS CARE CHRISTIAN INSTITUTE and may require further adjustment to be applicable to other institutions with different academic structures or operational requirements.

Definition of Terms

This section presents the conceptual and operational definitions of the key terms used throughout the study. Each term is defined within the context of the AEGIS system to ensure clarity and uniformity in interpretation. The definitions provided serve as a reference for readers to understand how specific words and phrases are applied in the discussions, analyses, and findings of this research.

AEGIS. The proposed system developed in this study, designed as a web-based AI-Enhanced Guidance and Integrated School Information System for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas Inc. that centralizes academic records, attendance, guidance services, and school announcements within a single secure environment accessible to students, parents, faculty, guidance counselors, and administrators.

Chatbot. A software application or web interface designed to simulate conversation with human users through text or voice interactions using artificial intelligence [8]. In this study, the chatbot refers to the AI-powered virtual assistant integrated into AEGIS that processes user queries, provides academic assistance,

assists guidance counselors in generating student assessments, and routes escalated concerns to the appropriate modules or personnel.

Collaborative Filtering. A recommendation technique used in recommender systems that predicts user preferences by analyzing patterns of behavior and interactions among users with similar profiles [9]. In this study, collaborative filtering refers to the feature of AEGIS that generates personalized academic recommendations for students based on their performance data and the patterns of students with comparable academic profiles.

Content Management System (CMS). A software application used for the creation, management, and modification of digital content without requiring specialized technical knowledge [10]. In this study, the CMS refers to the feature within the Administrator Module of AEGIS that enables school administrators to manage announcements, academic resources, and system-wide content, which also serves as the knowledge base for the AI Chatbot's responses.

Data Privacy Act of 2012 (Republic Act No. 10173). A Philippine law enacted to protect the fundamental right to privacy by governing the collection, processing, and storage of personal information in both government and private sector information and communications systems [11]. In this study, the Data Privacy Act of 2012 refers to the legal framework that governs the handling of all student

and institutional data within AEGIS, ensuring that the privacy rights of all users are upheld throughout the system's operation.

Decision Tree Algorithm. A non-parametric supervised machine learning algorithm used for both classification and regression tasks, structured hierarchically to represent decisions and their possible outcomes [12]. In this study, the Decision Tree algorithm refers to the analytical feature of AEGIS that processes imported academic and behavioral data to proactively identify students who may be at risk of academic difficulty and generate alerts for guidance counselors.

Guidance Module. A dedicated component of a school information system designed to support the documentation and management of student welfare concerns, consultation records, and counseling activities [13]. In this study, the Guidance Module refers to the section of AEGIS that enables guidance counselors to document consultation records, session logs, and behavioral incident reports, create and deploy student assessments, and receive AI-generated alerts for at-risk students.

Natural Language Processing (NLP). A subfield of computer science and artificial intelligence that uses machine learning to enable computers to understand, interpret, and communicate in human language [14]. In this study, NLP refers to the technology used by the AI Chatbot of AEGIS to process and classify user queries,

route escalated concerns to appropriate modules or personnel, and generate relevant responses based on the school's content and academic data.

Role-Based Access Control (RBAC). An access control mechanism that restricts system access based on the roles assigned to individual users, where permissions are granted according to the functions associated with each defined role [15]. In this study, RBAC refers to the security feature of AEGIS that ensures each user — whether student, parent, faculty, guidance counselor, or administrator — can only access the functions and information relevant to their designated role within the system.

School Information System. A management information system for educational institutions used to manage student data and support communication between students, parents, teachers, and administrators, covering functions such as grade management, attendance tracking, scheduling, and reporting [16]. In this study, the school information system refers to AEGIS as a whole, which integrates academic record management, guidance services, stakeholder communication, and AI-powered features within a single organized platform.

Web-Based System. A software application accessed through a web browser that operates over the internet without requiring local installation on a user's device [17]. In this study, web-based system refers to the nature of AEGIS as a platform

accessible through any browser by all registered users of HIS CARE CHRISTIAN INSTITUTE, provided that a stable internet connection is available.

CHAPTER 2

REVIEW OF RELATED SYSTEMS

This chapter provides an overview of the technical background and related systems that served as references for the development of AEGIS. It also outlines the software development methodology that guided the implementation of the project.

Technical Background

The development of AEGIS: An AI-Enhanced Guidance and Integrated School Information System is grounded on several important technical concepts that focus on improving school information management, enabling early identification of at-risk students, and delivering timely, accessible support to all stakeholders within the school community. These concepts are not limited to programming alone. They also concern how school data is collected, organized, analyzed, and acted upon in a way that is practical for students, parents, faculty, guidance counselors, and administrators of HIS CARE CHRISTIAN INSTITUTE of Santa Maria Sto Tomas, Batangas Inc. This section discusses the key technical foundations of the system and explains how these concepts are applied in addressing the real challenges identified in the study.

One of the foundational concepts applied in this study is web-based system design. Unlike native mobile applications that require installation and are limited to specific operating systems, a web-based platform can be accessed through any modern browser on any device, whether it is a laptop, desktop, tablet, or smartphone. This design decision directly addresses the varied technological backgrounds of the school's stakeholders, ensuring that AEGIS remains accessible without requiring additional software. The frontend of AEGIS is built using React, a JavaScript library developed by Meta for building fast and interactive user interfaces. React enables the development of a Single Page Application (SPA) architecture, where the interface updates dynamically without requiring full page reloads, resulting in a smoother and more responsive experience for all user roles. Each role, namely student, parent, faculty, guidance counselor, and administrator, is presented with a distinct portal tailored to their specific functions upon login.

To serve the frontend and manage incoming requests efficiently, AEGIS uses Nginx as its web server and reverse proxy. Nginx handles the routing of HTTPS requests from client browsers to the appropriate backend services, serves the React build as static assets, and supports HTTP/2 for faster communication between the client and server. This setup ensures that the platform remains stable, performant, and capable of handling multiple concurrent users from different roles without significant degradation in response time.

The backend of AEGIS is developed using Django, a high-level Python web framework known for its security features, scalability, and clean architectural design. Django handles the core application logic of the system, including user authentication, request processing, data validation, and coordination between the different modules. Communication between the React frontend and the Django backend is facilitated through a REST API Gateway built with Django REST Framework, which structures all data exchanges using the REST/JSON protocol over HTTPS. This separation of frontend and backend layers makes the system modular, maintainable, and scalable over time.

User authentication and access control in AEGIS are managed through JSON Web Tokens (JWT) combined with Role-Based Access Control (RBAC). When a user logs in, the system issues a JWT that encodes their identity and assigned role. Subsequent requests from the client include this token, allowing the backend to verify the user's identity and enforce role-specific access restrictions without maintaining server-side session state. This approach ensures that students, parents, faculty, guidance counselors, and administrators can only access the features and data appropriate to their role, protecting sensitive school records from unauthorized access.

The database layer of AEGIS is powered by PostgreSQL, a relational database management system chosen for its reliability, data integrity, and support

for complex queries. AEGIS organizes its data across multiple specialized databases: a Student DB for student profiles and enrollment data, an Academic DB for grades, schedules, and activities, an Attendance DB for attendance records, a Guidance DB for consultation logs, behavioral notes, and assessments, and an AI and Analytics DB for reports, forecasts, and at-risk student data. This structured separation allows the backend to retrieve and update records efficiently across multiple user sessions while ensuring that dashboards and reports always reflect the most current information available.

A key technical concept integrated into AEGIS is data import management. Rather than building a live integration with the school's existing Learning Management System, AEGIS supports the upload and processing of CSV and Excel files exported from the school. These files are mapped to the corresponding student, grade, and attendance records within the system, enabling data synchronization without requiring changes to the school's existing infrastructure. This approach makes AEGIS practical to deploy in institutions that are already used without the need for complex API integrations.

One of the most technically significant components of AEGIS is its Integration Layer, which houses the AI and analytics capabilities of the system. The AI and Analytics Engine incorporate Natural Language Processing for chatbot query understanding and classification, a Decision Tree Algorithm for identifying

at-risk students based on academic performance patterns, and Collaborative Filtering for generating personalized academic resource recommendations. The Decision Tree Algorithm analyzes imported academic data to detect patterns associated with academic risk, such as declining grades, low attendance, or a combination of performance indicators, and generates alerts that are sent to guidance counselors for timely follow-up. Collaborative Filtering, on the other hand, identifies students with similar academic profiles and uses those patterns to suggest relevant learning resources and intervention strategies, providing a personalized layer of support within the platform.

The AI Chatbot Engine in AEGIS is built using Botpress or Rasa as the conversational platform, integrated with the Gemini API for advanced natural language understanding. The chatbot serves all five user roles and is capable of processing queries, classifying their nature and urgency using NLP-based topic classification, routing concerns to appropriate modules or personnel, suggesting academic resources, and assisting guidance counselors in generating assessment questions. Unlike a passive FAQ tool, the chatbot in AEGIS is connected to the AI and Analytics Engine, allowing it to escalate flagged queries, such as those indicating emotional distress or academic urgency, directly to the guidance counselor's module for professional follow-up.

Complementing the chatbot is the Notification Service, which handles system-generated alerts and email reports. When a student is flagged as at risk by the Decision Tree model, when a budget alert is triggered, or when a new announcement is posted, the Notification Service ensures that the relevant stakeholders receive real-time alerts both within the platform and through email. This keeps all users informed and enables timely responses without requiring them to actively monitor the system at all times.

The Content Management System (CMS) service in AEGIS is implemented using a white-label CMS or Strapi, an open-source headless CMS that allows administrators to manage school announcements, news, and resources through a flexible and user-friendly interface. Strapi integrates with the Django backend through the REST API Gateway, ensuring that content updates are reflected across all user portals in real time.

Security is treated as a dedicated layer in the AEGIS architecture rather than an afterthought. The Security Layer implements JWT-based authentication, RBAC for role-specific access enforcement, HTTPS/TLS encryption for all data in transit, and input validation to prevent injection attacks and malformed data from reaching the backend. An Audit Trail component continuously logs all system-wide activities and administrative actions, recording who accessed or modified data, when, and what changes were made. This audit mechanism supports compliance with the Data

Privacy Act of 2012 (RA 10173) and enables administrators to detect and respond to unauthorized access or anomalous system behavior.

The infrastructure supporting AEGIS consists of a web and mobile server, a database server, backup and recovery mechanisms, and file and media storage. These components ensure that the system remains operational, recoverable in the event of failure, and capable of storing all necessary files and media assets generated throughout the platform's use. At the topmost layer, the Operations and Maintenance Layer includes system monitoring tools, logging services, a deployment pipeline for continuous updates, and administrative tools for managing the platform post-deployment. Together, these infrastructure and maintenance components ensure that AEGIS remains reliable, secure, and continuously improving over time.

In summary, the technical foundation of AEGIS integrates web-based system design using React and Django, efficient request routing through Nginx, secure authentication via JWT and RBAC, structured data management using PostgreSQL, data import through CSV and Excel processing, AI-powered analytics using a Decision Tree Algorithm and Collaborative Filtering, natural language chatbot support through Botpress or Rasa with Gemini API integration, real-time notifications, CMS-driven content management, a dedicated security and audit layer, and a robust infrastructure and maintenance stack. These concepts work

together to address the real-world challenges of school information management at HIS CARE CHRISTIAN INSTITUTE of Santa Maria Sto Tomas, Batangas Inc., providing a system that is not only technically sound but genuinely useful and accessible to every stakeholder it is designed to serve.

System Architecture

The development of AEGIS: An AI-Enhanced Guidance and Integrated School Information System is grounded on several important technical concepts that focus on improving school information management, enabling early identification of at-risk students, and delivering timely, accessible support to all stakeholders within the school community. These concepts are not limited to programming alone. They also concern how school data is collected, organized, analyzed, and acted upon in a way that is practical for students, parents, faculty, guidance counselors, and administrators of HIS CARE CHRISTIAN INSTITUTE of Santa Maria Sto Tomas, Batangas Inc. This section discusses the key technical foundations of the system and explains how these concepts are applied in addressing the real challenges identified in the study.

One of the foundational concepts applied in this study is web-based system design. Unlike native mobile applications that require installation and are limited to specific operating systems, a web-based platform can be accessed through any modern browser on any device, whether it is a laptop, desktop, tablet, or

smartphone. This design decision directly addresses the varied technological backgrounds of the school's stakeholders, ensuring that AEGIS remains accessible without requiring additional software. The frontend of AEGIS is built using React, a JavaScript library developed by Meta for building fast and interactive user interfaces. React enables the development of a Single Page Application (SPA) architecture, where the interface updates dynamically without requiring full page reloads, resulting in a smoother and more responsive experience for all user roles. Each role, namely student, parent, faculty, guidance counselor, and administrator, is presented with a distinct portal tailored to their specific functions upon login.

To serve the frontend and manage incoming requests efficiently, AEGIS uses Nginx as its web server and reverse proxy. Nginx handles the routing of HTTPS requests from client browsers to the appropriate backend services, serves the React build as static assets, and supports HTTP/2 for faster communication between the client and server. This setup ensures that the platform remains stable, performant, and capable of handling multiple concurrent users from different roles without significant degradation in response time.

The backend of AEGIS is developed using Django, a high-level Python web framework known for its security features, scalability, and clean architectural design. Django handles the core application logic of the system, including user authentication, request processing, data validation, and coordination between the

different modules. Communication between the React frontend and the Django backend is facilitated through a REST API Gateway built with Django REST Framework, which structures all data exchanges using the REST/JSON protocol over HTTPS. This separation of frontend and backend layers makes the system modular, maintainable, and scalable over time.

User authentication and access control in AEGIS are managed through JSON Web Tokens (JWT) combined with Role-Based Access Control (RBAC). When a user logs in, the system issues a JWT that encodes their identity and assigned role. Subsequent requests from the client include this token, allowing the backend to verify the user's identity and enforce role-specific access restrictions without maintaining server-side session state. This approach ensures that students, parents, faculty, guidance counselors, and administrators can only access the features and data appropriate to their role, protecting sensitive school records from unauthorized access.

The database layer of AEGIS is powered by PostgreSQL, a relational database management system chosen for its reliability, data integrity, and support for complex queries. AEGIS organizes its data across multiple specialized databases: a Student DB for student profiles and enrollment data, an Academic DB for grades, schedules, and activities, an Attendance DB for attendance records, a Guidance DB for consultation logs, behavioral notes, and assessments, and an AI

and Analytics DB for reports, forecasts, and at-risk student data. This structured separation allows the backend to retrieve and update records efficiently across multiple user sessions while ensuring that dashboards and reports always reflect the most current information available.

A key technical concept integrated into AEGIS is data import management. Rather than building a live integration with the school's existing Learning Management System, AEGIS supports the upload and processing of CSV and Excel files exported from the school. These files are mapped to the corresponding student, grade, and attendance records within the system, enabling data synchronization without requiring changes to the school's existing infrastructure. This approach makes AEGIS practical to deploy in institutions that already use without the need for complex API integrations.

One of the most technically significant components of AEGIS is its Integration Layer, which houses the AI and analytics capabilities of the system. The AI and Analytics Engine incorporates Natural Language Processing for chatbot query understanding and classification, a Decision Tree Algorithm for identifying at-risk students based on academic performance patterns, and Collaborative Filtering for generating personalized academic resource recommendations. The Decision Tree Algorithm analyzes imported academic data to detect patterns associated with academic risk, such as declining grades, low attendance, or a

combination of performance indicators, and generates alerts that are sent to guidance counselors for timely follow-up. Collaborative Filtering, on the other hand, identifies students with similar academic profiles and uses those patterns to suggest relevant learning resources and intervention strategies, providing a personalized layer of support within the platform.

The AI Chatbot Engine in AEGIS is built using Botpress or Rasa as the conversational platform, integrated with the Gemini API for advanced natural language understanding. The chatbot serves all five user roles and is capable of processing queries, classifying their nature and urgency using NLP-based topic classification, routing concerns to appropriate modules or personnel, suggesting academic resources, and assisting guidance counselors in generating assessment questions. Unlike a passive FAQ tool, the chatbot in AEGIS is connected to the AI and Analytics Engine, allowing it to escalate flagged queries, such as those indicating emotional distress or academic urgency, directly to the guidance counselor's module for professional follow-up.

Complementing the chatbot is the Notification Service, which handles system-generated alerts and email reports. When a student is flagged as at risk by the Decision Tree model, when a budget alert is triggered, or when a new announcement is posted, the Notification Service ensures that the relevant stakeholders receive real-time alerts both within the platform and through email.

This keeps all users informed and enables timely responses without requiring them to actively monitor the system at all times.

The Content Management System (CMS) service in AEGIS is implemented using a white-label CMS or Strapi, an open-source headless CMS that allows administrators to manage school announcements, news, and resources through a flexible and user-friendly interface. Strapi integrates with the Django backend through the REST API Gateway, ensuring that content updates are reflected across all user portals in real time.

Security is treated as a dedicated layer in the AEGIS architecture rather than an afterthought. The Security Layer implements JWT-based authentication, RBAC for role-specific access enforcement, HTTPS/TLS encryption for all data in transit, and input validation to prevent injection attacks and malformed data from reaching the backend. An Audit Trail component continuously logs all system-wide activities and administrative actions, recording who accessed or modified data, when, and what changes were made. This audit mechanism supports compliance with the Data Privacy Act of 2012 (RA 10173) and enables administrators to detect and respond to unauthorized access or anomalous system behavior.

The infrastructure supporting AEGIS consists of a web and mobile server, a database server, backup and recovery mechanisms, and file and media storage. These components ensure that the system remains operational, recoverable in the

event of failure, and capable of storing all necessary files and media assets generated throughout the platform's use. At the topmost layer, the Operations and Maintenance Layer includes system monitoring tools, logging services, a deployment pipeline for continuous updates, and administrative tools for managing the platform post-deployment. Together, these infrastructure and maintenance components ensure that AEGIS remains reliable, secure, and continuously improving over time.

In summary, the technical foundation of AEGIS integrates web-based system design using React and Django, efficient request routing through Nginx, secure authentication via JWT and RBAC, structured data management using PostgreSQL, data import through CSV and Excel processing, AI-powered analytics using a Decision Tree Algorithm and Collaborative Filtering, natural language chatbot support through Botpress or Rasa with Gemini API integration, real-time notifications, CMS-driven content management, a dedicated security and audit layer, and a robust infrastructure and maintenance stack. These concepts work together to address the real-world challenges of school information management atHIS CARE CHRISTIAN INSTITUTE od Sta. Maria Sto. Tomas Btangas, providing a system that is not only technically sound but genuinely useful and accessible to every stakeholder it is designed to serve.

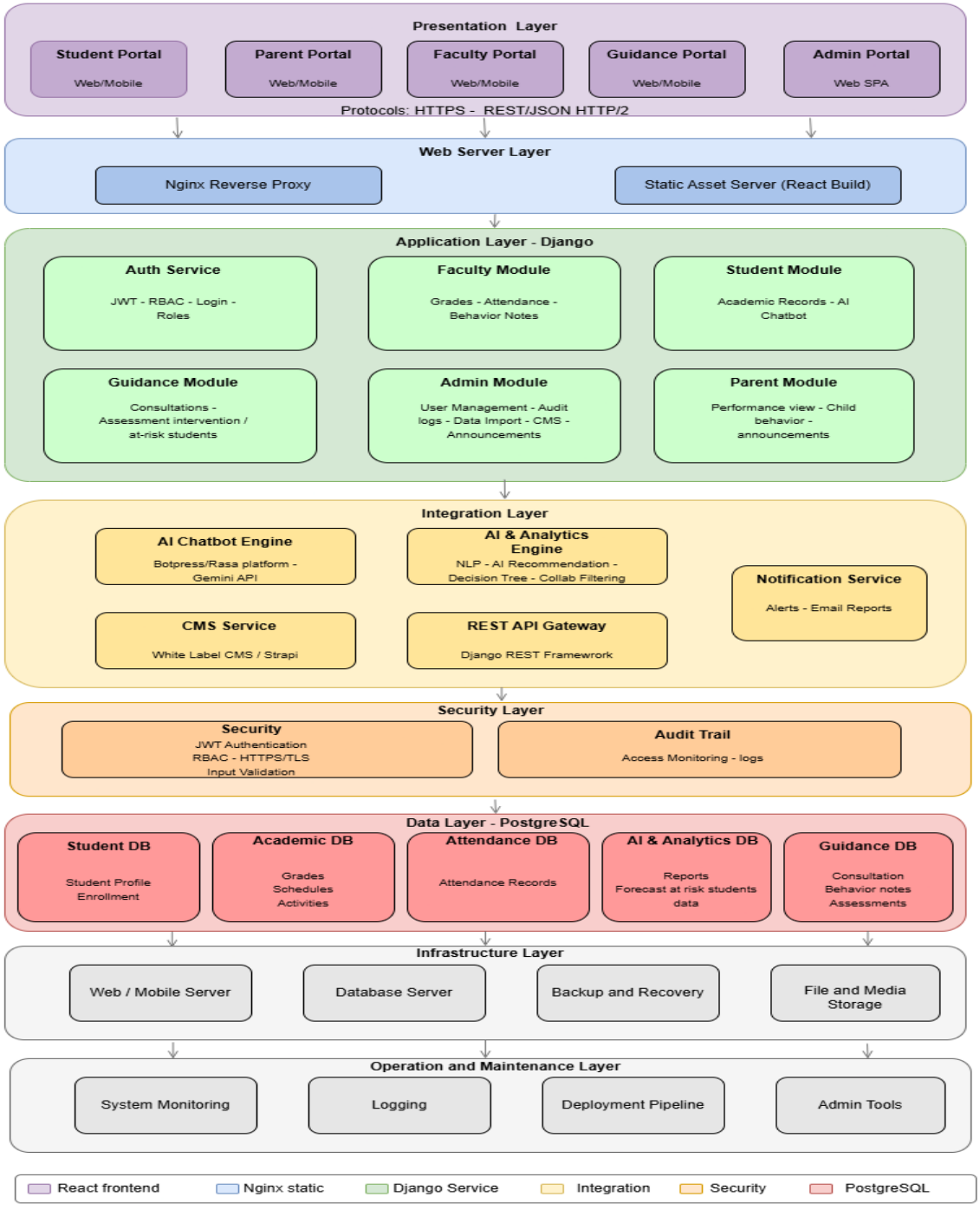


Figure 1. System Architecture Diagram

Related Systems

This section presents local and foreign systems related to the development of AEGIS: AI-Enhanced Guidance and Integrated School Information System. The reviewed systems provide insights into the implementation of school information management, academic monitoring, communication platforms, and artificial intelligence-assisted educational technologies. These systems serve as references in identifying effective approaches and existing limitations relevant to the proposed system.

Bation et al., designed and implemented an e-School System for a community college in Northern Mindanao, Philippines. The system centralized student records, enrollment, grading, scheduling, and administrative operations into a single web-based platform. The researchers found that the implementation of a centralized information system improved the efficiency of record management, reduced administrative workload, and increased the accessibility of institutional information among stakeholders [19].

The study is relevant to the development of AEGIS because both systems focus on improving school operations through centralized information management. Similar to the e-School System, AEGIS consolidates academic records, attendance, schedules, announcements, and guidance services within a unified platform accessible to administrators, faculty members, students, and parents. However,

while the system developed by Tugano et al. primarily focuses on administrative management and record organization, AEGIS extends these functionalities by integrating artificial intelligence features such as academic risk classification, recommendation systems, and an NLP-assisted chatbot designed to support guidance and intervention services.

Salvador, Ramos, and Villanueva developed a web-based student monitoring and attendance management system for secondary schools in the Philippines. The researchers addressed issues related to delayed attendance reporting, inconsistent student monitoring, and difficulties in communicating with parents. Their findings showed that the system improved the speed of attendance tracking, enhanced parent notifications, and reduced manual documentation processes within the institution [20].

The study supports the development of AEGIS because both systems emphasize real-time monitoring and improved communication between schools and stakeholders. Similar to the proposed system, the platform developed by Salvador et al. allows school personnel to manage student information digitally and provide timely updates to parents. However, AEGIS provides a broader and more integrated solution by combining attendance monitoring with academic performance tracking, behavioral documentation, guidance services, and predictive analytics for identifying academically at-risk students.

Santos and Rivera developed a web-based guidance management system intended to improve documentation and monitoring processes within school guidance offices. The researchers highlighted the difficulties encountered in maintaining paper-based counseling records, including limited accessibility, inconsistent documentation, and delays in retrieving student case histories. Their findings revealed that digitizing counseling records improved organization, confidentiality, and efficiency in generating reports for interventions and follow-up activities [21].

This study is relevant to AEGIS because both systems aim to improve guidance-related services through digital technologies. Similar to the guidance management system, AEGIS incorporates digital consultation records, behavioral incident documentation, and intervention tracking for guidance counselors. However, unlike the system developed by Santos and Rivera, AEGIS integrates guidance services with broader academic monitoring functionalities, allowing counselors and faculty members to correlate behavioral and counseling records with grades, attendance, and risk classifications to support more comprehensive student interventions.

Dela Cruz et al, developed a parent-school communication portal designed to improve stakeholder engagement in private educational institutions in the Philippines. The system enabled parents to access announcements, student grades,

attendance reports, and school activities through a centralized online platform. The researchers found that the portal improved parental awareness and increased participation in school-related activities and student monitoring [22].

The study is relevant to the development of AEGIS because both systems recognize the importance of direct and timely communication between schools and parents. Similar to the communication portal, AEGIS provides parents with role-based access to announcements, schedules, attendance records, and academic information. However, AEGIS further enhances parental support by integrating predictive monitoring features and AI-assisted functionalities that help identify students who may require intervention before academic difficulties become severe.

Agada and Eze conducted a study on the role of Information and Communication Technology (ICT) in improving school record management systems. The researchers examined how traditional paper-based processes affect the efficiency of record keeping in educational institutions. Their findings revealed that schools relying on manual documentation commonly experience delays in retrieving records, data inconsistencies, duplication of files, and difficulties in maintaining accurate institutional information. The study further concluded that the implementation of ICT-based systems significantly improves record accessibility, organization, security, and overall administrative efficiency [23].

The study supports the development of AEGIS because both emphasize the importance of replacing fragmented and paper-based school processes with centralized digital systems. Similar to the study, AEGIS addresses inefficiencies in handling academic records, attendance, guidance documentation, and institutional announcements through an integrated platform. However, while the study focuses generally on ICT implementation in educational institutions, AEGIS further incorporates AI-assisted technologies such as predictive monitoring and NLP-based support functionalities tailored specifically for secondary school operations.

González-Sancho and Vincent-Lancrin discussed how fragmented educational information systems create inefficiencies in data management, record retrieval, and institutional decision-making. The authors explained that educational institutions that rely on disconnected files, spreadsheets, and manual documentation processes often encounter inconsistencies, duplicated records, and difficulties in accessing accurate information needed for timely decision-making. Their study emphasized that centralized educational information systems improve accessibility, consistency, and operational efficiency by reducing duplication and administrative burden [24].

The study is relevant to AEGIS because the proposed system addresses the same challenges identified in the literature, particularly the reliance on fragmented files, spreadsheets, and manual documentation processes within educational

institutions. Similar to the study, AEGIS promotes centralized record management to improve data organization and accessibility. However, while the study focuses broadly on the transformation of educational systems through information technologies, AEGIS applies these concepts specifically within a secondary school setting while integrating AI-assisted monitoring and guidance support functionalities.

Tugano, Dator, and Bation conducted a study on the design and implementation of an e-School System for a community college in Northern Mindanao, Philippines. The researchers developed a web-based platform that centralized enrollment, student records, grading, scheduling, and other administrative processes into a unified information system. Their findings showed that the implementation of the platform significantly improved administrative efficiency, reduced manual workloads, and enhanced accessibility to institutional records among stakeholders [25].

This study supports the development of AEGIS because both systems aim to improve school operations through centralized digital management. Similar to the study, AEGIS seeks to minimize administrative inefficiencies associated with manual documentation and disconnected systems. However, AEGIS further extends these functionalities by integrating predictive analytics, AI-assisted guidance support, and recommendation systems for academic intervention.

PowerSchool is a widely used student information system implemented in schools across the United States. The platform provides centralized management of student records, grades, attendance, schedules, communication tools, and analytics for administrators, teachers, students, and parents. Studies examining the implementation of PowerSchool reported improvements in administrative efficiency, accessibility of academic records, and communication among school stakeholders [26].

The system is relevant to the development of AEGIS because both systems focus on centralized school information management and stakeholder accessibility. Similar to PowerSchool, AEGIS provides role-based portals for administrators, faculty members, students, parents, and guidance counselors. However, while PowerSchool primarily focuses on large-scale institutional information management, AEGIS specifically incorporates AI-assisted guidance support, academic risk classification, and recommendation functionalities tailored to the operational needs of a private secondary school environment.

Dougiamas and Taylor discussed the development and implementation of Moodle, an open-source learning management system widely used internationally for course management, academic monitoring, and communication between educators and students. Their study revealed that Moodle improves accessibility to

educational resources, supports student engagement, and simplifies academic management through centralized digital tools [27].

The study is relevant to AEGIS because both systems utilize web-based technologies to improve access to academic information and educational support services. Similar to Moodle, AEGIS promotes organized record management and communication between stakeholders through a centralized platform. However, while Moodle primarily focuses on online learning delivery and classroom management, AEGIS emphasizes institutional record management, guidance services, student risk monitoring, and AI-assisted intervention support.

Researchers and institutions implementing Ellucian Banner examined how enterprise-level academic information systems improve institutional operations through centralized data management. The system integrates student records, enrollment management, academic tracking, financial services, and institutional reporting within a single platform. Findings highlighted its effectiveness in improving institutional efficiency, data accessibility, and decision-making processes within educational institutions [28].

The system supports the development of AEGIS because both platforms emphasize centralized and organized information management for educational institutions. Similar to Ellucian Banner, AEGIS consolidates records and administrative functions into a single accessible environment. However, AEGIS

differs by focusing specifically on secondary education support services and by integrating artificial intelligence functionalities such as decision-tree-based academic monitoring and NLP-assisted concern classification.

Google developed Google Classroom as a cloud-based educational platform intended to facilitate communication, assignment management, and collaboration between teachers and students. Studies on the platform indicate that it improves accessibility, simplifies communication, and enhances digital learning experiences through integrated educational tools and centralized online access [29].

This system is relevant to AEGIS because both systems aim to improve communication and accessibility within educational environments. Similar to Google Classroom, AEGIS enables digital interaction between faculty members, students, and parents through centralized announcements and academic monitoring tools. However, unlike Google Classroom, which focuses mainly on classroom instruction and assignment distribution, AEGIS incorporates institutional record management, guidance support, academic analytics, and predictive monitoring functionalities designed for school-wide operational management.

Ouyang, Xu, and Cukurova conducted a comprehensive review on the application of data-driven artificial intelligence technologies in education. The researchers examined how AI systems are utilized for academic monitoring, predictive analytics, personalized learning, and educational decision-making. Their

findings revealed that AI-assisted technologies improve the ability of institutions to identify students requiring intervention through pattern recognition and data analysis. The study also emphasized that artificial intelligence enables schools to shift from reactive to proactive student support strategies [30].

The study strongly supports the development of AEGIS because the proposed system integrates artificial intelligence functionalities for academic monitoring and guidance assistance. Similar to the reviewed literature, AEGIS utilizes decision-tree analysis and recommendation techniques to identify academically at-risk students and provide intervention support. However, while the study broadly discusses AI applications in education, AEGIS implements these technologies specifically within a centralized secondary school information system.

Muraina, Aiyegbusi, and Abam conducted a study on the use of Decision Tree algorithms in predicting students' academic performance in advanced programming courses. The researchers analyzed student performance indicators and applied predictive classification techniques to identify learners who may require additional academic support. Their findings demonstrated that Decision Tree models effectively classify students according to performance levels and provide reliable support for academic monitoring and intervention processes [31].

The study is relevant to AEGIS because both systems apply predictive analysis techniques for identifying academically at-risk students. Similar to the

study, AEGIS analyzes grades, attendance, and behavioral indicators to classify student risk levels. However, while Muraina et al. focused primarily on predictive performance classification within a single academic subject, AEGIS integrates broader institutional records and guidance-related indicators to support holistic student monitoring and intervention.

Caldarini et al., conducted a literature survey on recent advances in chatbot technologies and Natural Language Processing applications. The study explored how chatbot systems improve accessibility to information, automate communication processes, and enhance user support services within digital platforms. The findings showed that NLP-based chatbots significantly improve response efficiency, streamline communication, and increase user engagement when integrated properly within organizational systems [32].

This study supports the development of AEGIS because the proposed system integrates an NLP-assisted chatbot for handling student concerns and institutional inquiries. Similar to the reviewed study, AEGIS utilizes Natural Language Processing for intent detection and concern classification. However, unlike general-purpose chatbots discussed in the literature, the chatbot in AEGIS is designed specifically for controlled educational environments and provides institution-approved recommendations and intervention support aligned with school policies.

The reviewed local and foreign systems demonstrate that digital school management platforms significantly improve information accessibility, communication, administrative efficiency, and student monitoring within educational institutions. Existing systems commonly focus on centralized records management, communication tools, and learning support services. However, most systems either emphasize administrative operations or instructional delivery without fully integrating academic risk monitoring, AI-assisted guidance support, and stakeholder-centered intervention functionalities within a single platform.

Taken together, these gaps support the development of AEGIS: AI-Enhanced Guidance and Integrated School Information System. Unlike existing systems that focus primarily on isolated school management functions, AEGIS combines centralized records management, academic monitoring, guidance services, artificial intelligence-assisted support, recommendation systems, and predictive analysis within a unified web-based environment specifically designed for secondary school operations.

CHAPTER 3
DESIGN AND METHODOLOGY

This chapter will discuss the design and methodologies to be used by the developers in creating the system. It will also present the diagrams and data that have be gathered to guide the development process and support the completion of the system. Furthermore, this chapter will explain the type of research to be used in the study and how it will be applied in developing the system.

Data Gathering

The proponents used a qualitative method to collect essential data for the proposed system. Qualitative research focused on non-numerical information that helped understand people's opinions, experiences, and daily routines. For AEGIS, this approach was employed to learn about the specific needs, difficulties, and existing practices of the school principal and faculty administrator of HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas when it came to managing student records, monitoring academic performance, handling guidance services, and communicating with parents and students.

The primary technique for data collection was semi-structured interviews with the school's principal. This method gave the researchers the chance to ask predetermined questions while still being able to explore unexpected answers more deeply. The interview covered six major areas: the school's current profile and

existing systems, administrative challenges during grading and record management, the process of identifying and supporting at-risk students, guidance office documentation practices, how parents and students currently access academic information, and the school's overall readiness to adopt a web-based digital system. Participants were encouraged to share their real experiences, frustrations, and ideas about what an improved system should offer.

The people who participated in this data gathering was the school principal of HIS CARE CHRISTIAN INSTITUTE, who was selected because the principal is the most directly involved in managing school operations, academic records, and student support services. The researchers arranged the interview schedule directly with the institution's administration. By speaking with key personnel responsible for overseeing day-to-day school functions, the proponents were able to gather a clear and grounded picture of the institution's specific challenges and operational gaps. This helped ensure that AEGIS was built around what the school community truly needed, increasing the likelihood that the system would be effectively adopted and consistently used once deployed.

Project Concept

The proposed system focuses on providing a digital solution to help HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas manage its academic operations, student support services, and school communications more

efficiently. The proponents developed this idea not only to address the recurring problems of fragmented record management and delayed information sharing but also to introduce a technology that is genuinely responsive to the realities of a small private high school in a local setting. While school information systems already exist in the market, most are built for larger institutions, require costly licensing, and lack features specifically designed for the combined needs of academic monitoring, guidance services, and proactive identification of at-risk students. AEGIS is built as a web-based platform accessible through standard browsers, making it practical for use in a school environment where devices range from desktop computers to mobile phones.

For academic record management, teachers and class advisers will be able to encode grades, attendance, and behavioral notes directly into a centralized system rather than maintaining separate files and spreadsheets. The system will consolidate these records in real time, significantly reducing the one to two days of manual reconciliation that faculty members currently spend before every card-giving day. Administrators will also be able to generate official reports and access updated student information at any time without the need to gather documents from multiple sources. Role-based access control will ensure that each type of user, administrator, faculty, guidance counselor, parent, and student can only view and manage information relevant to their role within the institution.

AEGIS will also incorporate an AI-driven academic monitoring component powered by a Decision Tree algorithm. This feature will analyze each student's grades, attendance, and behavioral records to classify them into risk levels — low, moderate, or high. Rather than waiting for a teacher or adviser to manually observe and report a struggling student, the system will proactively flag students who show early signs of academic difficulty and notify the appropriate personnel for timely intervention. This shifts the institution's approach to student support from reactive to anticipatory, allowing guidance counselors and faculty to respond before a student's difficulties become more severe.

For guidance services, the system will provide a structured digital environment where consultation records, session logs, and behavioral incident reports can be documented and tracked over time. Guidance counselors will be able to monitor the progress of ongoing interventions, maintain organized student histories, and generate reports needed for institutional planning. Additionally, AEGIS will include an AI-assisted chatbot that uses Natural Language Processing to classify student concerns by intent — whether academic, behavioral, emotional, or general — and respond with counselor-approved resources or recommendations. For serious concerns, the system will escalate notifications to the guidance counselor for direct follow-up.

Parents and students will also be given their own access portals within the system. Parents will be able to view their child's grades, attendance records, and any academic or behavioral concerns raised by the school, removing the current dependence on reactive communication through phone calls or personal visits. Students will similarly have access to their own academic records and school announcements without needing to go to the school physically. The system will also support school-wide announcement management, allowing administrators to publish and manage institutional updates through a single organized channel.

Through these features, AEGIS aims to provide a practical and intelligent platform that reduces administrative burden, strengthens communication between the school and its stakeholders, and enables a more proactive and data-driven approach to student support. By bringing together centralized record management, AI-assisted academic monitoring, structured guidance services, and role-based information access into one unified system, AEGIS hopes to become an essential part of daily operations at HIS CARE CHRISTIAN INSTITUTE.

Development Method

A development methodology serves as a guide in organizing and managing the processes involved in building a system. It provides a structured approach that helps the developers plan, design, implement, and evaluate the proposed system effectively. This section discusses the methodology adopted by the proponents in

developing the proposed system and explains how each phase contributes to the overall system development.

Figure 2 shows that the AEGIS project will adopt the Agile Software Development methodology. Agile is a flexible and iterative development approach that emphasizes collaboration, continuous improvement, and regular feedback from stakeholders. Through this approach, the system will be developed in smaller stages or iterations, allowing the developers to continuously improve the system based on the needs and feedback of the intended users. This methodology also enables the developers to quickly identify issues, implement necessary changes, and adapt system features throughout the development process.

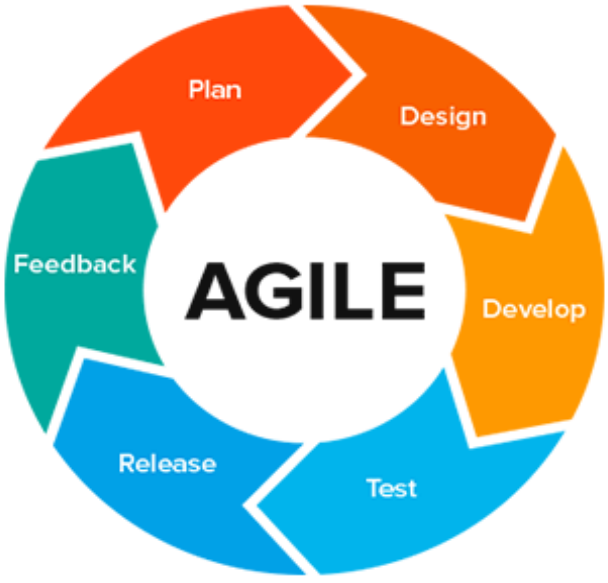


Figure 2. Agile Methodology

The first stage of the development process is the planning phase. In this phase, the proponents will determine the main objectives of the study and identify the stakeholders who will benefit from the system, including the administrators, faculty members, guidance counselors, parents, and students of HIS CARE CHRISTIAN INSTITUTE. The overall scope of the project, timeline of activities, and major system features will also be outlined during this stage.

After the planning stage, the requirement gathering phase will be conducted. The proponents will collect information from the school's principal and faculty administrator to understand the problems they encounter in academic record management, guidance documentation, and school communication. These collected insights will be translated into system requirements that will serve as the foundation for the system's functionality and features.

The next stage is the system design phase. In this phase, the structure of the system will be planned, including the database organization, user interface layout, and system components needed for the web application. The design will focus on creating an accessible and easy-to-use interface that allows users to manage academic records, monitor student performance, and access guidance and communication services efficiently.

Following the design stage is the development phase, where the actual coding and system construction will take place. The developers will build the features of the system step by step while continuously checking if each component functions properly. This iterative development approach allows improvements to be made throughout the process rather than waiting until the end of the project.

Once the system features have been developed, testing will be conducted to ensure that the system performs accurately and reliably. Different testing methods will be used to verify that each function operates as intended. Any identified errors or system issues will be corrected before the system is prepared for implementation.

The final stage is deployment and evaluation. During this phase, the system will be introduced to the intended users, and guidance will be provided to help them understand how to navigate and operate the web application. Feedback from users will be collected and analyzed to determine the effectiveness of the system and identify possible improvements.

By applying the Agile development methodology, the proponents aim to develop a system that effectively supports academic record management, student monitoring, and school communication. This approach ensures that the system remains adaptable to the needs of its users while continuously improving its functionality and performance.

Development Approach

The development of AEGIS: An AI-Enhanced Guidance and Integrated School Information System for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas follows a Top-Down Approach in building the system. This approach begins with identifying the overall goals and structure of the system before gradually moving toward the implementation of detailed components and functionalities. By starting from the general concept and progressing into more specific system elements, the proponents can ensure that the development process remains organized and aligned with the objectives of the study.

During the planning stage, the proponents will gather relevant information from the school's principal and faculty administrator of HIS CARE CHRISTIAN INSTITUTE. Interviews will be conducted to identify their current challenges in academic record management, student monitoring, guidance documentation, and school communication. This phase will also involve determining the scope of the project, identifying the stakeholders, and establishing the necessary resources needed for the development of the system.

In the design stage, the overall architecture of the system will be established. This includes defining the structure of the web application, designing the user interface, and planning the database structure that will store all necessary system data. The design will prioritize simplicity and usability to ensure that users can

easily navigate the system and access essential features such as managing academic records, monitoring student performance, and accessing guidance and communication services.

During the development stage, the individual components of the system will be built and integrated to form a functional web application. The front-end and back-end elements of the system will be developed and connected to ensure that all system functions operate properly. Throughout this phase, continuous testing will be conducted to detect and resolve any technical issues, errors, or performance concerns that may affect the system's operation.

The deployment phase will involve preparing the system for actual use by the intended users. This includes setting up the web application within the school's environment, providing basic guidance to users on how to navigate the system, and ensuring that the system is accessible and operational. After deployment, the system will be monitored to evaluate its performance and to provide technical support if necessary.

Figure X illustrates the development approach that will be used in creating the proposed system. The figure demonstrates how the development process progresses from general planning toward more detailed implementation stages. Through this structured approach, the proponents aim to produce a reliable and efficient system that meets the needs and expectations of its users while supporting

better academic record management, student monitoring, and school communication at HIS CARE CHRISTIAN INSTITUTE.

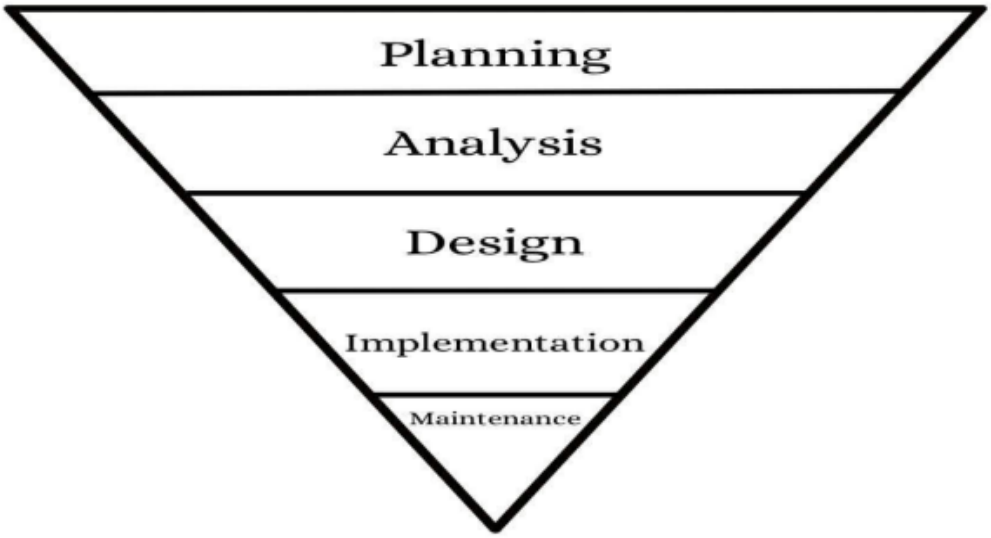


Figure 3. Top-Down Approach

System Analysis and Design

The system analysis and design discuss the approach and methods that will be used in the development of AEGIS: AI-Enhanced Guidance and Integrated School Information System for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas. The system will focus on providing a web-based solution designed to support academic record management, student monitoring, guidance services, and school communication within a single centralized platform. Visual representations such as context diagrams and data flow diagrams will be

utilized to clearly illustrate how data will move within the system and how users will interact with its components.

The design of the system will emphasize a user-centered approach, ensuring that the interface will be simple, intuitive, and suitable for users with varying levels of technical familiarity. Since the system will be accessed by different types of users including administrators, faculty members, guidance counselors, parents, and students, the interface will be designed with clear navigation, organized layouts, and role-specific dashboards to ensure that each user can efficiently access and manage the information relevant to their role.

In addition, the system will be designed with data security and access control as a priority. Role-based access control will be implemented to ensure that sensitive student records, guidance consultation logs, and behavioral reports can only be accessed by authorized users. Authentication and session management will be handled through Laravel Breeze to maintain secure and reliable user access throughout the system.

Furthermore, the system will integrate key features such as grade and attendance management, AI-driven academic monitoring, guidance documentation, an AI-assisted chatbot, batch file processing, and role-based information portals into a single platform. These features will be interconnected to provide a seamless workflow, allowing administrators, faculty members, and guidance counselors to

maintain accurate records and ensure that timely information reaches parents and students. The design will also consider system performance and scalability to accommodate multiple users, academic records, and institutional data. Efficient data handling and structured database management will be implemented to ensure reliable system performance across different devices and browsers.

By combining user-centered design, secure access control, and integrated functionality, the system aims to provide a practical and effective solution that supports the real-world needs of HIS CARE CHRISTIAN INSTITUTE in managing its academic operations and student support services.

Context Flow Diagram

Figure 4 presents the data flow diagram for the Guidance Counselor of the AEGIS system. The process begins with the Guidance Counselor logging into the system through the Guidance Login process, where authentication is validated and login credentials are retrieved from the guidance_tbl data store. Once authenticated, the Guidance Counselor can access the View Guidance Dashboard, which retrieves and displays aggregated data needed for managing student consultation records and monitoring at-risk students.

From the dashboard, the Guidance Counselor can proceed to the Manage Consultation Records process, where consultation data is documented and maintained within the system. The consultation data is then passed to the Receiving

AI Generated Alert for At-Risk Students process, where AI analytics data is retrieved from the AIanalytics_tbl data store. This process allows the Guidance Counselor to receive automated alerts generated by the system's AI component identifying students who may be academically at risk.

The AI analytics data is then forwarded to the Creating Student Assessment with AI process, where assessment data is stored in the assessment_tbl data store. The assessment data is subsequently passed to the Deploy Student Assessment process, which makes the assessment available for student use. The assessment results are then forwarded to the Generating Student Assessment Report process, where student assessment reports are stored in the report_tbl data store. Finally, the Export Student Assessment Report process delivers the student assessment output file back to the Guidance Counselor, allowing them to review, export, and use the reports for intervention planning and institutional documentation.

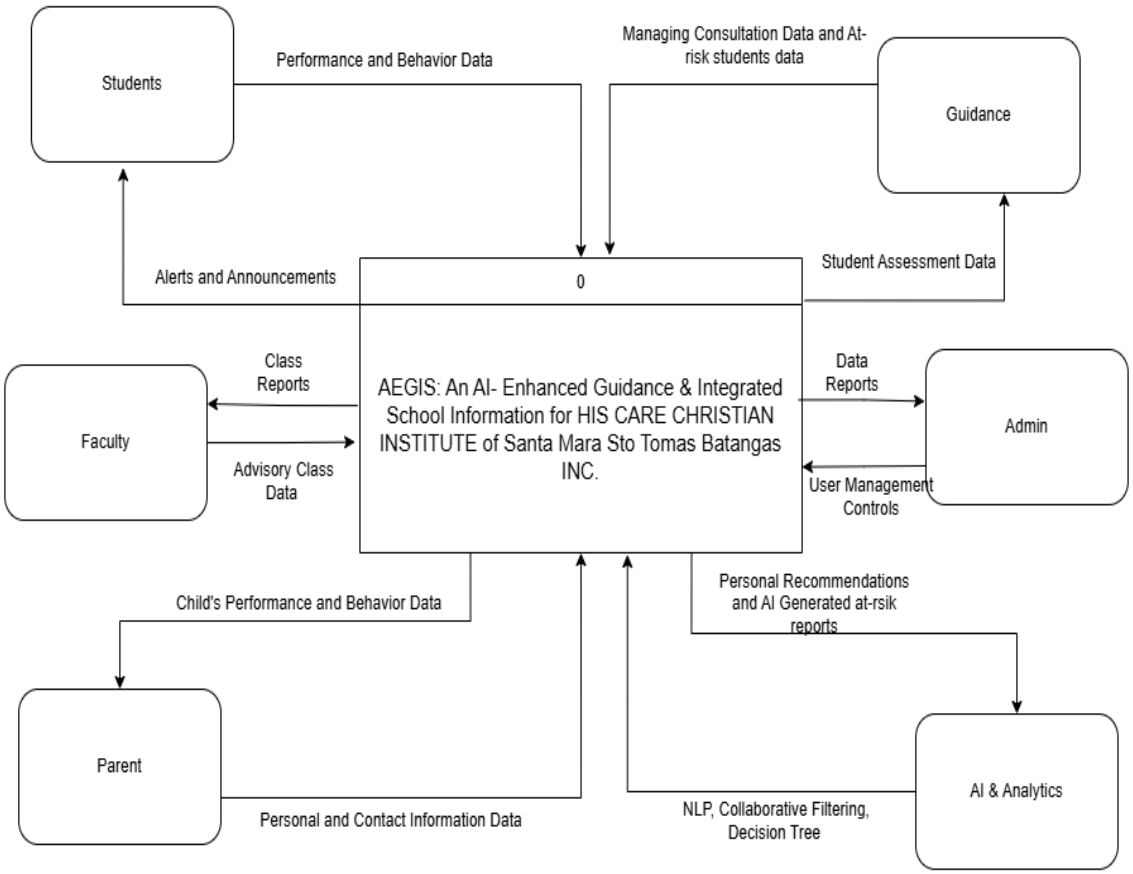


Figure 4. Context Flow Diagram of the AEGIS

Data Flow Diagram

A data flow diagram provides a visual representation of how information moves through a system. It shows the path of data from external entities to internal processes, then to data stores, and finally back to the users. This type of diagram helps developers understand the boundaries of the system and how different components interact with one another.

Figure 5 presents the data flow diagram for the Admin of the AEGIS system.

The process begins with the Admin logging into the system through the Login Admin process, where authentication is validated using the admin_tbl data store. Once authenticated, the Admin can access the View Admin Dashboard, which retrieves and displays administrative data needed for managing school operations.

From the dashboard, the Admin can proceed to the Importing School Files process, where academic records, attendance data, and guidance record data are retrieved from the academic_tbl, attendance_tbl, and guidancerecord_tbl data stores respectively. The imported data is then passed to the Manage User Accounts process, where student, faculty, guidance, and parent accounts are managed using data from the student_tbl, faculty_tbl, guidance_tbl, and parent_tbl data stores. Role access configurations are then forwarded to the System Configuration and Control process, which consolidates aggregated data and passes it to the School Announcement Creation process.

The School Announcement Creation process stores announcement data in the announcement_tbl and forwards it to the Announcement Deployment process. The deployed announcements are then returned to the Admin as Announcement Data,

completing the administrative workflow within the system.

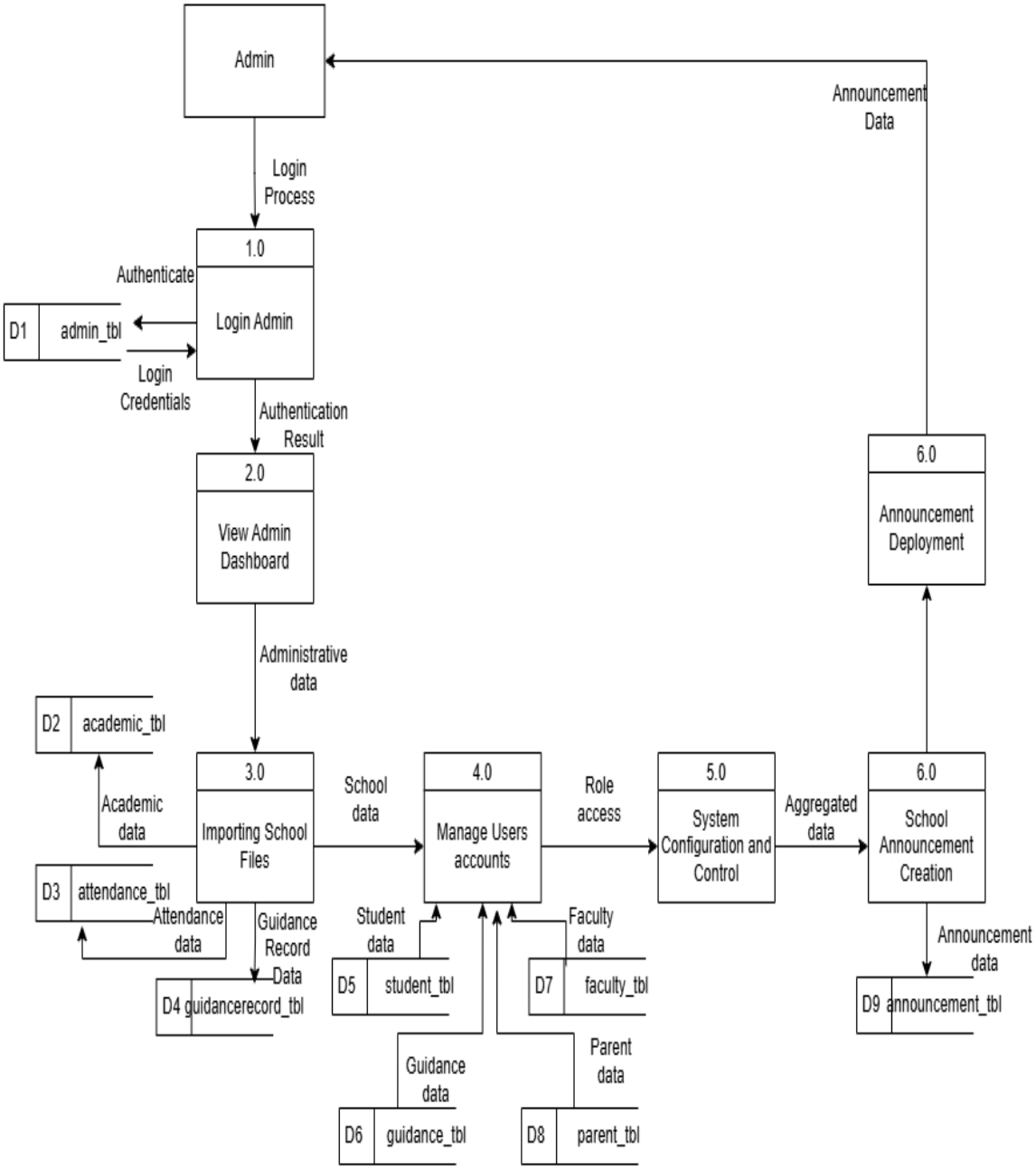


Figure 5. Data Flow Diagram of the Admin of AEGIS

Figure 6 presents the data flow diagram for the Faculty of the AEGIS system. The process begins with the Faculty logging into the system through the Faculty Login process, where authentication is validated using the faculty_tbl data store. Once authenticated, the Faculty can access the View Faculty Dashboard, which retrieves and displays aggregated data needed for managing class records and academic responsibilities.

From the dashboard, the Faculty can proceed to the Uploading/Importing Class Records process, where academic data and attendance data are retrieved from the academic_tbl and attendance_tbl data stores respectively. The uploaded data is then passed to the Manage Class Records process, where faculty members can organize, update, and maintain the class records of their students. The managed records are then forwarded to the Generating Reports process, where report data is stored in the report_tbl data store and the generated report output is sent to the Admin for review and institutional use.

The system also includes the Alerts and Announcement Creation process, which receives announcement and alerts information and delivers the finalized alerts and announcements back to the Faculty, keeping them informed of important school updates and notifications.

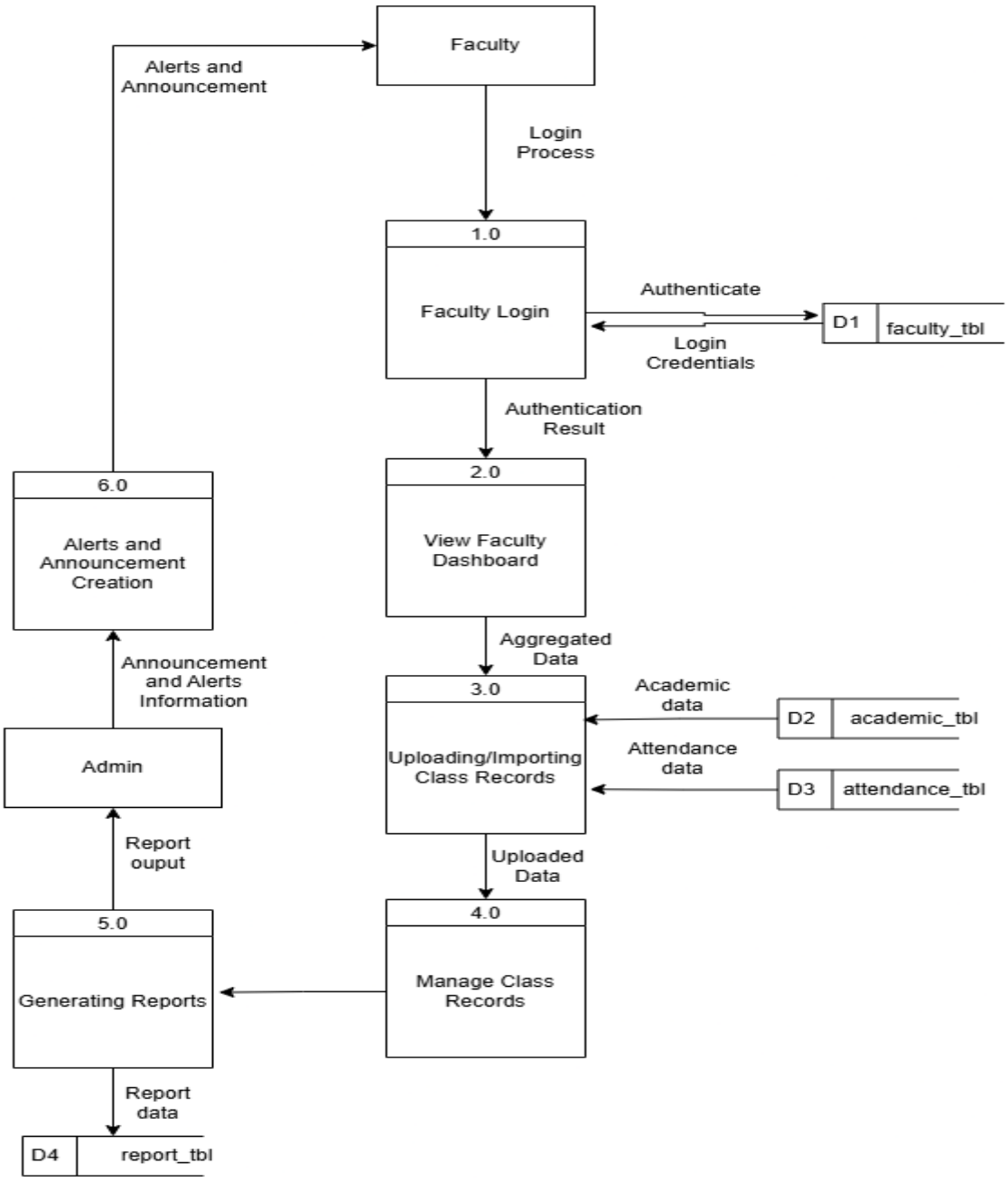


Figure 6. Data Flow Diagram of The Faculty of AEGIS

Figure 7 presents the data flow diagram for the Student of the AEGIS system. The process begins with the Student logging into the system through the Student Login process, where authentication is validated and login credentials are retrieved from the students_tbl data store. Once authenticated, the Student can access the View Student Dashboard, which retrieves and displays aggregated data relevant to the student's academic profile.

From the dashboard, the Student can proceed to the Monitor Academic Records process, where academic data is retrieved from the academic_tbl data store and monitored records are displayed for the student's review. The monitored data is then passed to the Taking Assessment process, where assessment result data is stored in the assessment_tbl data store. Reports generated from the assessment are forwarded to the Alerts and Announcement Creation process, which delivers alerts and notifications back to the Student to keep them informed of important academic updates and school announcements.

The system also includes the Interacting AI Chatbot process, where the Student can submit raw data or queries, and the chatbot returns recommendation data based on the student's submitted concerns. This allows the Student to receive AI-assisted guidance and academic support directly through the system.

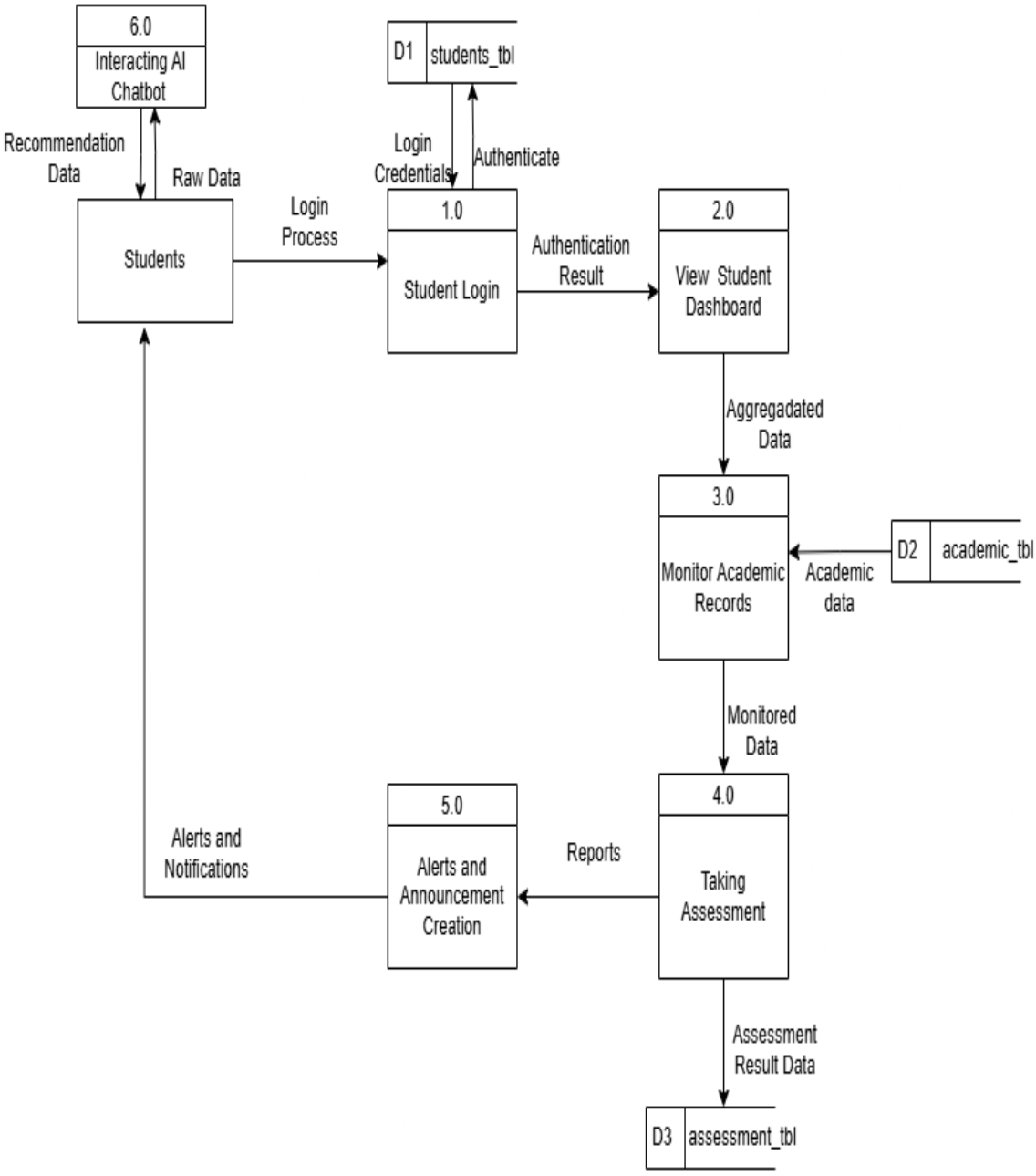


Figure 7. Data Flow Diagram of The Students of AEGIS

Figure 8 presents the data flow diagram for the Parent of the AEGIS system. The process begins with the Parent logging into the system through the Parent Login process, where authentication is validated and login credentials are retrieved from the parent_tbl data store. Once authenticated, the Parent can access the View Parent Dashboard, which retrieves and displays aggregated data relevant to the parent's account and their child's academic profile.

From the dashboard, the Parent can proceed to the View/Monitor Child's Performance and Behavior process, where academic data, attendance data, and guidance data are retrieved from the academic_tbl, attendance_tbl, and guidancerecord_tbl data stores respectively. This process consolidates all relevant information about the child's academic standing, attendance records, and guidance-related concerns into a single accessible view for the parent.

The performance data gathered from this process is then forwarded to the Generate Performance Report process, where report data is stored in the report_tbl data store. The generated report output is then returned directly to the Parent, allowing them to view a comprehensive summary of their child's academic performance and behavior without needing to visit the school physically.

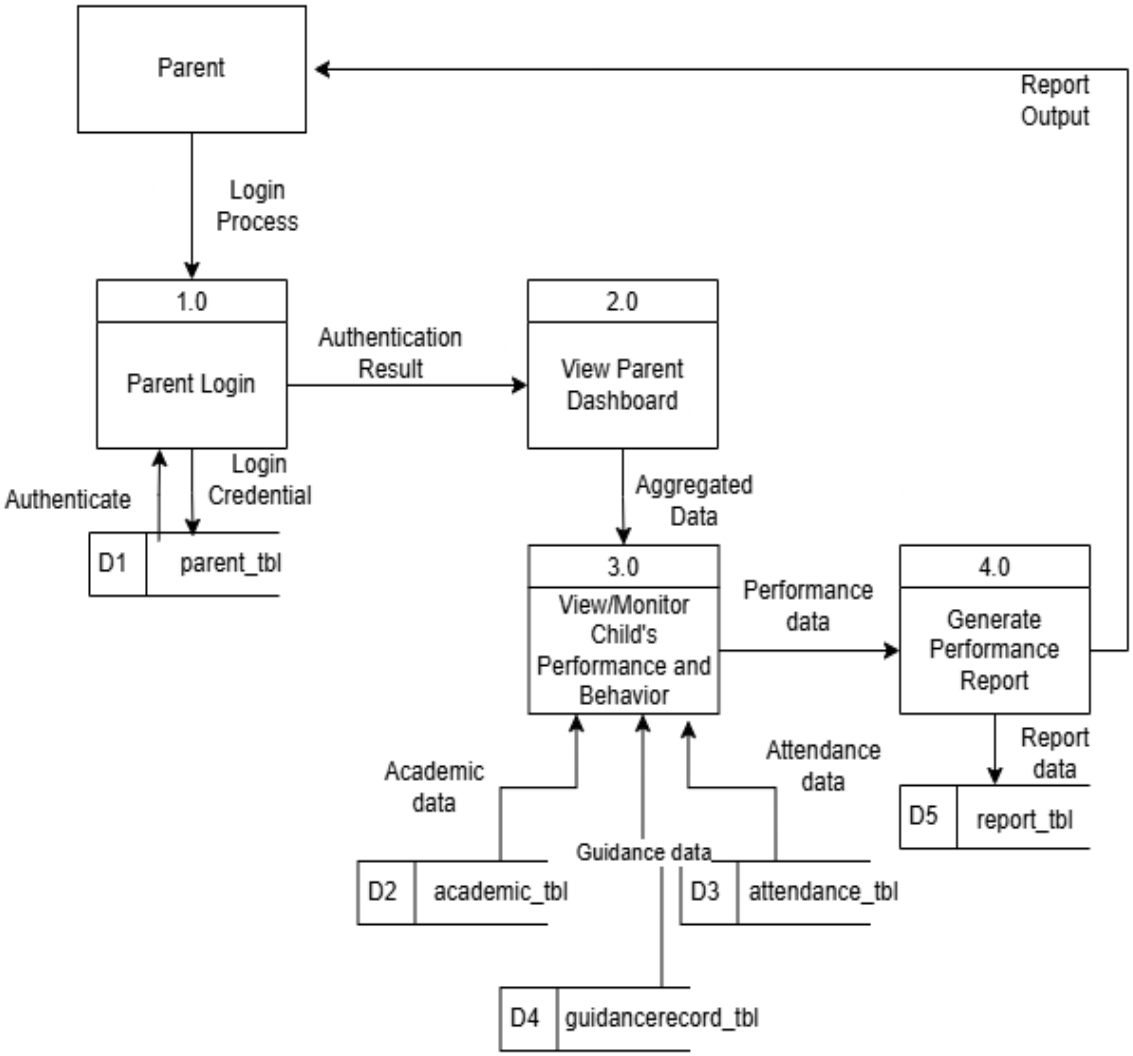


Figure 8. Data Flow Diagram of The Parent of AEGIS

Figure 9 presents the data flow diagram for the Guidance Counselor of the AEGIS system. The process begins with the Guidance Counselor logging into the system through the Guidance Login process, where authentication is validated and

login credentials are retrieved from the guidance_tbl data store. Once authenticated, the Guidance Counselor can access the View Guidance Dashboard, which retrieves and displays aggregated data needed for managing student consultation records and monitoring at-risk students.

From the dashboard, the Guidance Counselor can proceed to the Manage Consultation Records process, where consultation data is documented and maintained within the system. The consultation data is then passed to the Receiving AI Generated Alert for At-Risk Students process, where AI analytics data is retrieved from the AIanalytics_tbl data store. This process allows the Guidance Counselor to receive automated alerts generated by the system's AI component identifying students who may be academically at risk.

The AI analytics data is then forwarded to the Creating Student Assessment with AI process, where assessment data is stored in the assessment_tbl data store. The assessment data is subsequently passed to the Deploy Student Assessment process, which makes the assessment available for student use. The assessment results are then forwarded to the Generating Student Assessment Report process, where student assessment reports are stored in the report_tbl data store. Finally, the Export Student Assessment Report process delivers the student assessment output file back to the Guidance Counselor, allowing them to review, export, and use the reports for intervention planning and institutional documentation.

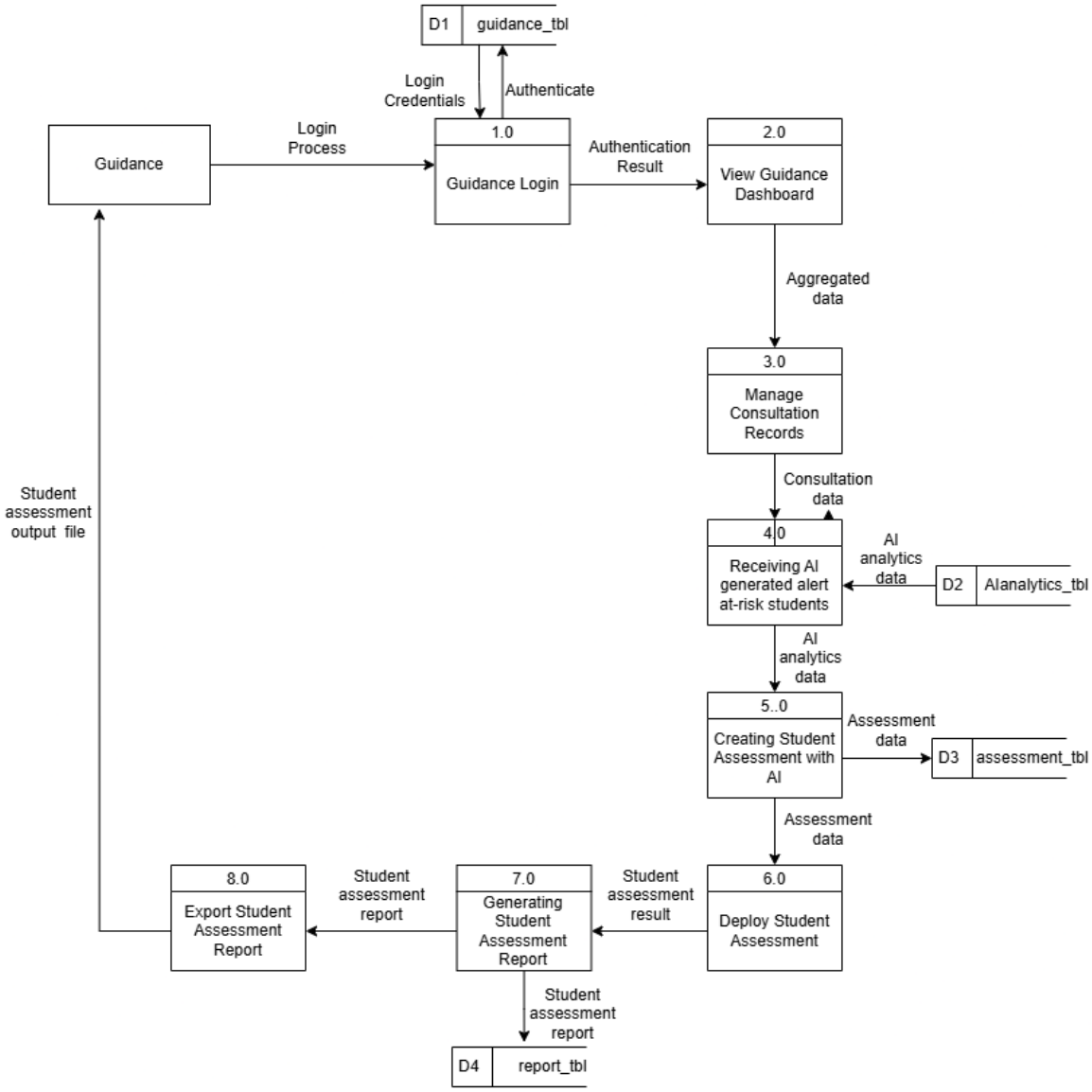


Figure 9. Data Flow Diagram of The Guidance of AEGIS

Functional Requirement

Functional requirements specify the features and operations that AEGIS must be able to perform. These requirements describe the expected behavior of the

system in response to user inputs and ensure that all necessary functionalities are covered.

This section outlines the functional requirements for AEGIS, a web-based platform developed to consolidate academic records, student monitoring, guidance services, and school communications for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas. The requirements cover the essential capabilities of the system, including user authentication and role-based access control, academic record and attendance management, batch file importing, guidance consultation documentation, AI-driven at-risk student detection, AI-assisted chatbot interaction, collaborative filtering recommendations, announcement management, report generation, and administrative user account management.

Table 1 presents the functional requirements of AEGIS, organized by unique identifier and category. Each requirement defines a specific system action that must be implemented and validated during the testing phase.

Table 1.

Functional Requirements

ID	Characteristics	Description
FR-001	Authentication	The system shall allow users to register.

Table 1.

Functional Requirements (cont'd)

FR-002	Authentication	The system shall assign a role (Student) to each user upon registration or account creation.
FR-003	Authentication	The system shall assign a role (Parent) to each user upon registration or account creation.
FR-004	Authentication	The system shall assign a role (Guidance Councilor) to each user upon registration or account creation.
FR-005	Authentication	The system shall assign a role (Faculty) to each user upon registration or account creation.
FR-006	Authentication	The system shall assign a role (Administrator) to each user upon registration or account creation.
FR-007	Faculty Module	The system shall allow faculty to create and manage advisory class records including student profiles.
FR-008	Faculty Module	The system shall allow faculty to create and manage advisory class records including attendance.
FR-009	Faculty Module	The system shall allow faculty to create and manage advisory class imcluding behavioral notes.
FR-010	Faculty Module	The system shall allow faculty to encode student grades .
FR-011	Faculty Module	The system shall allow faculty to update student grades.

Table 1.

Functional Requirements (cont'd)

FR-012	Faculty Module	The system shall support batch uploading of grades data through CSV or Excel files.
FR-013	Faculty Module	The system shall support batch uploading of attendance data through CSV or Excel files
FR-014	Faculty Module	The system shall display a consolidated one-page summary of advisory class data for each adviser.
FR-015	Faculty Module	The system shall allow faculty to generate and export subject performance.
FR-016	Faculty Module	The system shall allow faculty to generate and export advisory summaries.
FR-017	Student Module	The system shall allow students to view their grades.
FR-018	Student Module	The system shall allow students to view their class schedule.
FR-019	Student Module	The system shall allow students to view their attendance records.
FR-020	Student Module	The system shall deliver school announcements and real-time notifications to students regarding grades.
FR-021	Student Module	The system shall deliver school announcements and real-time notifications to students regarding attendance.
FR-022	Student Module	The system shall deliver school announcements and real-time notifications to students regarding concerns.
FR-023	Student Module	The system shall allow students to take AI-generated assessments deployed by the guidance counselor.
FR-024	Parent Module	The system shall allow parents to view their child's academic performanc in real time.

Table 1.

Functional Requirements (cont'd)

FR-025	Parent Module	The system shall allow parents to view their child's attendance in real time.
FR-026	Parent Module	The system shall allow parents to view their child's behavioral concerns in real time.
FR-027	Guidance Module	The system shall allow counselors to create student assessments with AI chatbot assistance for question generation.
FR-028	Guidance Module	The system shall allow counselors to deploy student assessments with AI chatbot assistance for question generation.
FR-029	Guidance Module	The system shall allow counselors to monitor student assessments with AI chatbot assistance for question generation.
FR-030	Guidance Module	The system shall generate AI-driven at-risk alerts for students based on academic performance patterns using a Decision Tree Algorithm.
FR-031	Guidance Module	The system shall flag and escalate AI Chatbot interactions that indicate student concerns requiring counselor follow-up.
FR-032	Guidance Module	The system shall allow guidance counselors to generate individual student guidance reports.
FR-33	Admin Module	The system shall provide a CMS for managing school announcements.
FR-034	Admin Module	The system shall provide a CMS for managing school system wide-setting
FR-035	Admin Module	The system shall provide a CMS for managing school resources.
FR-036	Admin Module	The system shall allow administrators to manage user accounts across all roles.

Table 1.

Functional Requirements (cont'd)

FR-037	Admin Module	The system shall support the upload and mapping of CSV/Excel files.
FR-038	Admin Module	The system shall maintain audit logs of all system-wide activities and administrative actions.
FR-039	Analytics	The system shall produce AI-driven recommendations using Collaborative Filtering.
FR-040	Analytics	The system shall produce AI-driven recommendations using Decision Tree algorithms.

Non-Functional Requirements

Non-functional requirements describe the quality attributes and performance standards that AEGIS must meet. These requirements define how the system behaves under various conditions rather than what specific functions it performs, ensuring that the system operates efficiently, securely, and reliably in a real school environment.

This section outlines the non-functional requirements for AEGIS, a web-based platform developed to consolidate academic records, student monitoring, guidance services, and school communications for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas. The requirements cover the essential quality attributes of the system, including performance and response time,

usability and accessibility across different devices and browsers, reliability and system availability, security and data privacy, maintainability, and scalability to accommodate multiple users and growing institutional data over time.

Table 2 presents the non-functional requirements of AEGIS, organized by unique identifier and category. Each requirement defines a specific quality standard that must be met and verified during the testing and evaluation phase.

Table 2.

Non-Functional Requirements

ID	Requirement Category	Description
NFR-001	Usability	The system shall present a consistent layout and navigation menu across all pages.
NFR-002	Usability	The system shall display clear error messages indicating what went wrong and how to correct it.
NFR-003	Usability	The system shall complete any user action, such as saving data or loading a page, within 5 seconds under normal network conditions.
NFR-004	Performance	The system shall support up to 500 concurrent users without degrading response time by more than 2 seconds.
NFR-005	Performance	The dashboard shall load and display all data within 5 seconds of login.
NFR-006	Data Integrity	The system shall prevent duplicate registration using the same email address.

Table 2.

Non-Functional Requirements (cont'd)

NFR-007	Reliability	The system shall produce consistent outputs from the Decision Tree Algorithm given the same input data.
NFR-008	Security	The system shall encrypt all user passwords using a hashing algorithm prior to storage.
NFR-009	Security	Students shall only be able to view and modify their own data.
NFR-010	Security	Administrators shall only be able to view student data in aggregated or anonymized form unless consent is given.
NFR-011	Security	The system shall enforce role-based access control to ensure users can only access data and features assigned to their role.
NFR-012	Availability	The system shall notify users at least 24 hours in advance of any scheduled maintenance.
NFR-013	Maintainability	The system shall be developed using modular code with separate files for frontend, backend, and database logic.
NFR-014	Compatibility	The system shall function correctly on the latest versions of Google Chrome, Mozilla Firefox, and Microsoft Edge browsers.
NFR-015	Compatibility	The system shall be responsive and usable on screen sizes from 320 pixels (mobile) to 1920 pixels (desktop).

User Design Interface

Figure [10] shows the Login and Register dashboard that consists of (1) "Email Textbox" where the user enters their registered email address to access their account, (2) "Password Textbox" where the user enters their confidential password as a masked input for security, (3) "Remember Me Checkbox" that allows the user to stay logged in on their device across sessions, (4) "Forgot Password Link" that redirects the user to a password recovery page to reset their credentials, (5) "Login Button" that submits the entered credentials to authenticate and grant the user access to the system, (6) "Sign Up Link" that navigates the user to the Register screen for new account creation, (7) "Name Textbox" where the new user enters their full name during registration, (8) "Email Textbox" where the new user provides their email address to be used as their account identifier (9) "Password Textbox" where the new user sets a password for their account, (10) "Confirm Password Textbox" where the user re-enters their password to verify it matches and prevent input errors, (11) "Sign Up Button" that submits the registration form to create a new user account in the system, and (12) "Login Link" that redirects already registered users back to the Login screen.

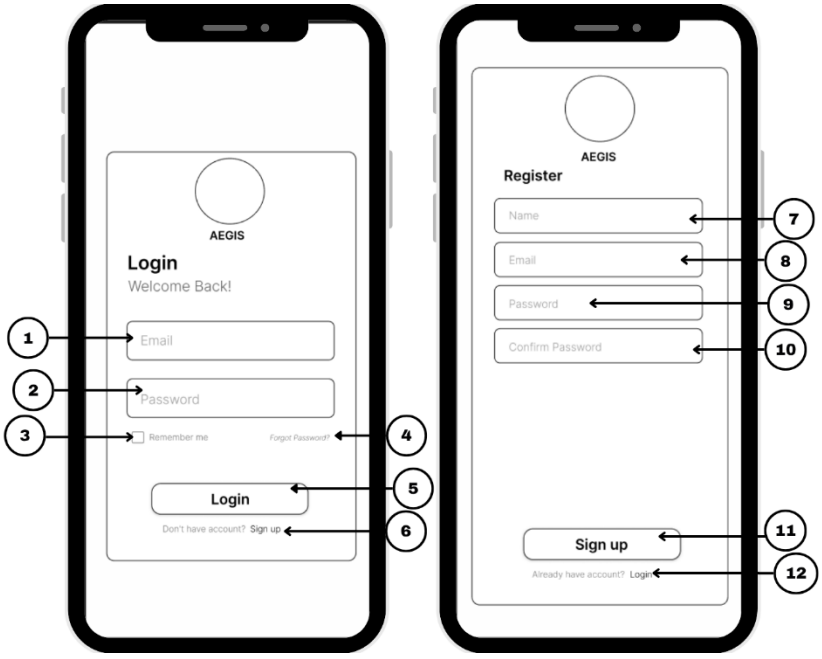


Figure [10]. Login Interface

1. **Email Textbox**
2. **Password Textbox**
3. **Remember Me Checkbox**
4. **Forgot Password Link**
5. **Login Button**
6. **Sign Up Link**
7. **Name Textbox**
8. **Email Textbox**
9. **Password Textbox**
10. **Confirm Password Textbox**
12. **Sign Up Button**
13. **Login Link**

Figure [11] shows the Home dashboard interface that consists of (1) "Hamburger Button" represented by a hamburger icon that opens the navigation drawer or sidebar menu, (2) "User Greeting Card" that displays the student's name and section such as "Hello, Juan – Grade 10 Einstein" along with a profile photo,

(3) "GPA Card" that shows the student's current Grade Point Average expressed as a percentage, (4) "Academic Progress Chart" that visually presents the student's GPA trend over the current quarter through a line graph plotted across months from September to January, (5) "Student Performance Section" that displays a list of subjects alongside their corresponding progress bars and numerical scores indicating the student's performance per subject, (6) "Notification Bell" represented by a bell icon that alerts the user of new notifications or announcements, (7) "Attendance Card" that displays the student's current attendance rate expressed as a percentage, (8) "Home Button" represented by a house icon that navigates the user back to the main dashboard, (9) "Academic Button" represented by a book icon that directs the user to the learning materials or library section, (10) "Reports Button" represented by a document icon that navigates the user to their academic reports or records, (11) "Profile Button" represented by a person icon that takes the user to their personal profile and account settings, and (12) "Announcements Section" that displays a preview list of recent school announcements such as exam schedules and upcoming events.

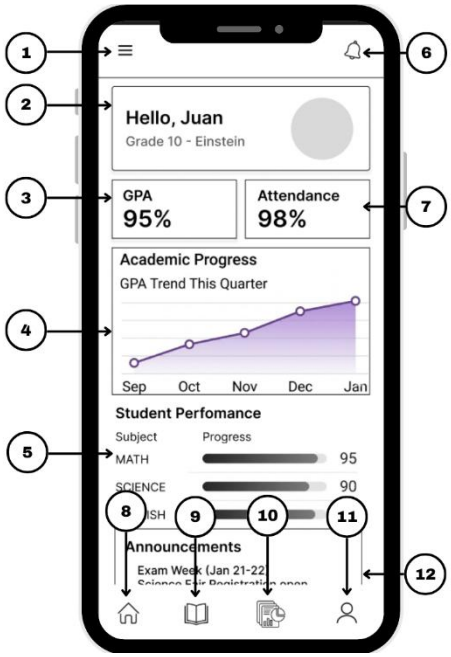


Figure 11. Home Dashboard Interface

1. **Hamburger Button**
2. **User Greeting Card**
3. **GPA Card**
4. **Academic Progress Chart**
5. **Student Performance Section**
6. **Notification Bell**
7. **Attendance Card**
8. **Home Button**
9. **Academic Button**
10. **Reports Button**
11. **Profile Button**
12. **Announcements Section**

Figure [12] shows the Academic Assessment interface that consists of (1) "Assessment Card" that displays the subject name, assessment title, and estimated duration for each available test, allowing the user to view the details of every subject assessment such as Mathematics – Algebra Readiness Test (10 Minutes), Science –

Science Learning Evaluation (15 Minutes), English – English Grammar Assessment (8 Minutes), and Araling Panlipunan – Philippine History Quiz (12 Minutes), and (2) "Start Assessment Button" that is present on each assessment card and allows the user to begin the corresponding subject assessment upon tapping.

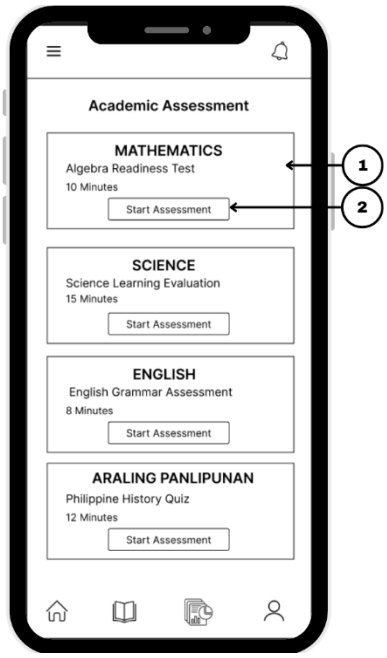


Figure [12]. Academic Assessment Interface

1. Assessment Card **2. Start Assessment**

Figure [13] shows the Analytics interface that consists of (1) "Academic Overview Card" that displays the student's attendance summary including the number of days present, late, and absent alongside a trend line graph that visualizes attendance or academic performance over time, (2) "Subject Performance Section" that presents each subject along with a progress bar and a corresponding numerical

score reflecting the student's current performance level per subject including Mathematics, Science, English, and Filipino, (3) "Assessment Performance Section" that lists the student's scores on previously taken assessments such as Algebra Test, Science Quiz, and Grammar Assessment each displayed with their respective scores out of ten, (4) "AI Academic Insights Section" that provides an AI-generated summary of the student's academic standing by highlighting strengths, areas that need improvement, and behavioral recommendations such as maintaining attendance consistency, and (5) "Learning Recommendations Section" that displays a personalized list of suggested study activities generated based on the student's performance data such as practicing algebra exercises, reviewing Science vocabulary, and continuing reading activities.

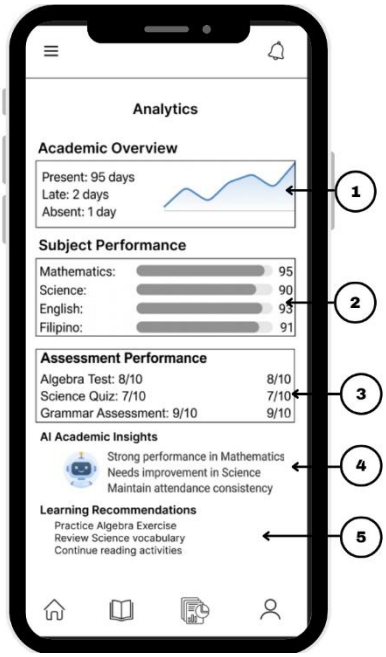


Figure [13]. Analytics Interface

1. **Academic Overview Card**
2. **Subject Performance Section**
3. **Assessment Performance Section**
4. **AI Academic Insights Section**
5. **Learning Recommendations Section**

Figure **[**14 **]** shows the Hamburger Button that consists of (1) "AI Assistant Button" that navigates the user to the AI-powered academic assistant feature where they can ask questions and receive personalized academic guidance, (2) "Announcements Button" that directs the user to the full list of school announcements and updates, and (3) "Settings Button" that takes the user to the application settings where they can manage their account preferences and configurations. The drawer also displays the student's profile photo and full name at the top, in this case showing "Juan Dela Cruz," serving as a profile header for the navigation panel.

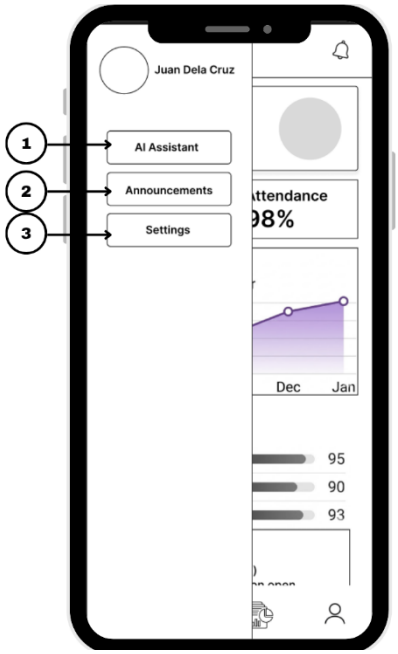


Figure [14]. Hamburger Menu Interface

1. **AI Assistant Button**
2. **Announcements Button**
3. **Settings Button**

Figure [15] shows the Profile interface that consists of (1) "Profile Header" that displays the student's profile photo, full name, email address, and current grade and section serving as the identity summary at the top of the screen, (2) "Profile Information Section" that presents the student's detailed personal information in a structured list format including Grade, Section, Email, Phone Number, and Address, and (3) "Edit Profile Button" that allows the user to modify and update their personal information and account details.

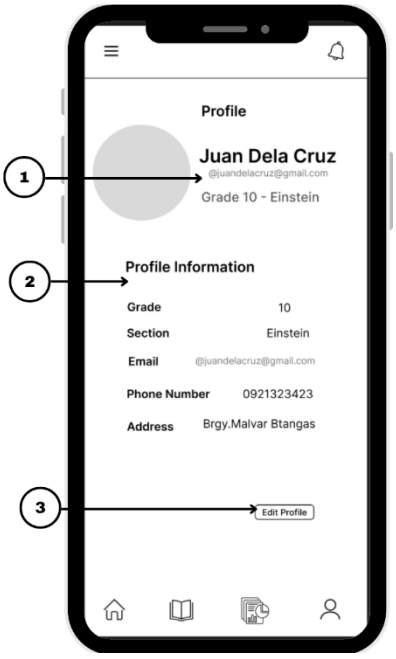


Figure [15]. Profile Interface

1. **Profile Header**
2. **Profile Information Section**
3. **Edit Profile Button**

Figure [16] shows the AI Chatbot interface that consists of (1) "Chat Display Area" that shows the conversation thread between the student and the AI assistant, displaying the AI's responses such as greetings and academic guidance messages related to lessons, grades, and school concerns, (2) "Message Input Textbox" that allows the user to type their question or message intended for the AI assistant, and (3) "Send Button" that submits the typed message to the AI assistant and triggers a response within the chat display area.

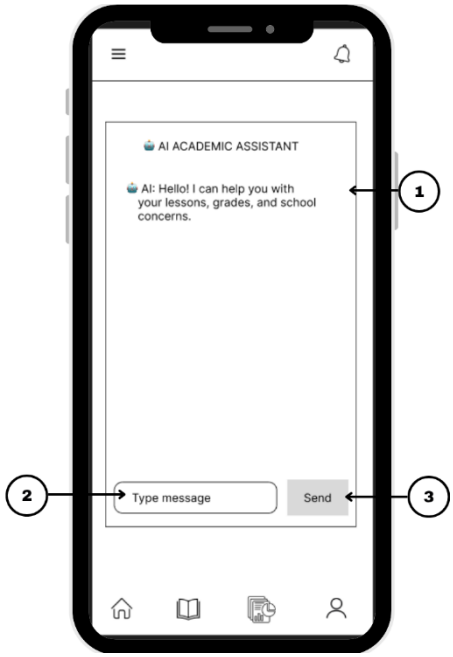


Figure [16]. AI Chatbot Interface

1. **Chat Display Area**
2. **Message Input Textbox**
3. **Send Button**

Figure [17] shows the Settings interface that consists of (1) "Search Bar" that allows the user to quickly search for a specific setting by typing keywords, (2) "Account Option" that navigates the user to account-related settings where they can manage their personal account details, (3) "Appearance Option" that directs the user to display and theme settings allowing them to customize the visual appearance of the application, (4) "Privacy & Security Option" that takes the user to privacy and security settings where they can manage permissions and account protection preferences, (5) "Help and Support Option" that redirects the user to the help center

or support resources for assistance with application concerns, (6) "About Option" that provides information about the application such as version details, developer information, and terms of use, and (7) "Logout Button" that allows the user to securely sign out of their account and return to the Login screen

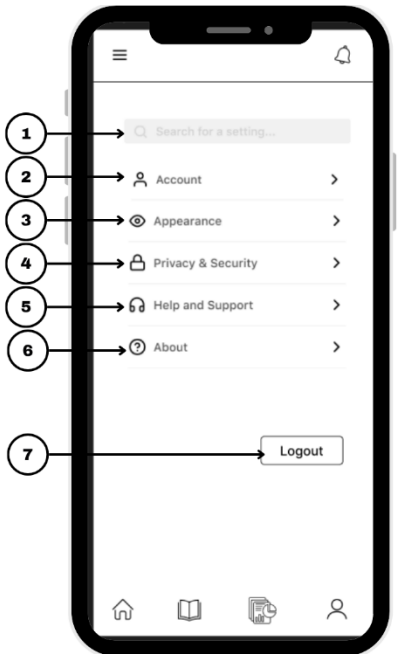


Figure [17]. Settings Interface

1. **Search Bar**
2. **Account Option**
3. **Appearance Option**
4. **Privacy & Security Option**
5. **Help and Support Option**
6. **About Option**
7. **Logout Button**

Figure [18] shows the Parent Dashboard interface that consists of (1) "Monitoring Header Card" that displays a welcome message for the parent along

with the name, grade, and section of the student being monitored, in this case showing "Welcome, Mrs. Juan – Monitoring: Juan Dela Cruz, Grade 10 – Einstein" alongside a profile photo, (2) "GPA Card" that shows the monitored student's current Grade Point Average expressed as a percentage, (3) "Announcements Section" that displays a preview list of recent school announcements and notices such as exam schedules, event registrations, and holiday notices, (4) "Academic Progress Chart" that visually presents the student's GPA trend over the current quarter through a line graph plotted across months from September to January, (5) "Student Performance Section" that displays a list of subjects alongside their corresponding progress bars and numerical scores reflecting the student's academic performance per subject, and (6) "Attendance Card" that shows the monitored student's current attendance rate expressed as a percentage.

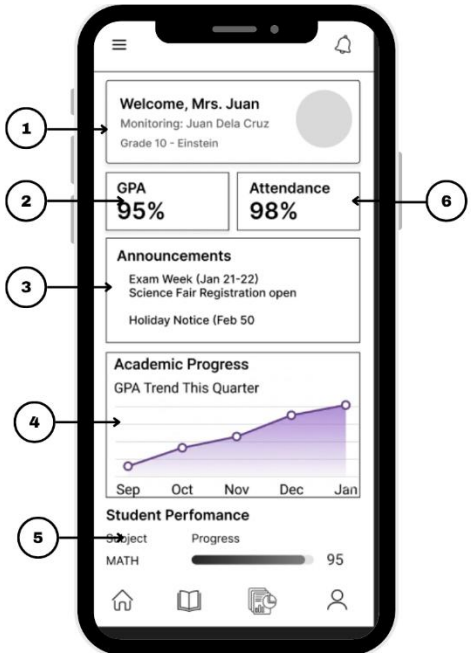


Figure [18]. Parent Dashboard Interface

1. **Monitoring Header Card**
2. **GPA Card**
3. **Announcements Section**
4. **Academic Progress Chart**
5. **Student Performance Section**
6. **Attendance Card**

Figure [19] shows the Student Information interface under the parent dashboard that consists of (1) "Student Details Card" that displays the student's complete basic information including full name, grade level, section, and assigned class adviser, (2) "Attendance Record Card" that presents a breakdown of the student's attendance showing the total number of days present, late, and absent for the current period, (3) "Behavioral Notes Card" that provides a summary of the student's observed classroom behavior and participation as recorded by the teacher,

such as being active in the classroom and participating in discussions, and (4) "Complete Grades Card" that displays the student's final or current grades across all subjects including Mathematics, Science, English, and Filipino with their corresponding numerical scores.

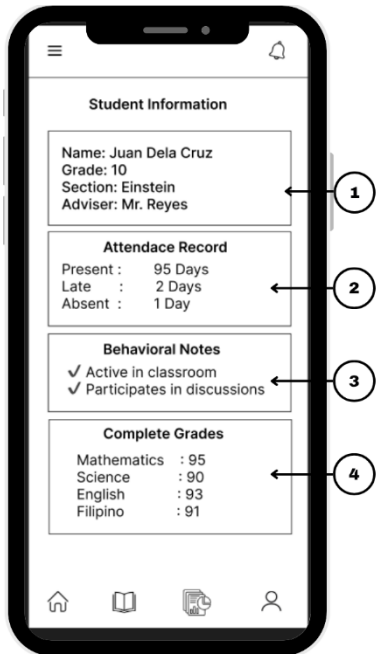


Figure [19]. Student Information Interface

1. **Student Details Card**
2. **Attendance Record Card**
3. **Behavioral Notes Card**
4. **Complete Grades Card**

Figure [20] shows the Parent Profile interface that consists of (1) "Profile Header" that displays the parent's profile photo, full name, email address, and relationship to the student such as "Maria Cruz – @mariacruz@gmail.com – Relationship: Mother" serving as the identity summary at the top of the screen, (2)

"Contact Information Section" that presents the parent's personal contact details in a structured format including email address, phone number, and home address, and (3) "Linked Student Section" that displays the name, grade level, and section of the student associated with the parent's account, in this case showing "Juan Dela Cruz – Grade 10 – Einstein" indicating the parent-student account connection within the system.

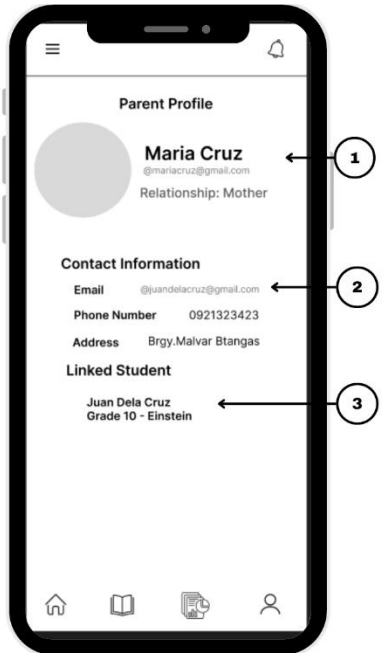


Figure [20]. Parent Profile Interface

1. **Profile Header**
2. **Contact Information Section**
3. **Linked Student Section**

Figure [21] shows the Web Login interface for Faculty, Guidance, and Admin users that consists of (1) "Email Textbox" where the authorized user enters their registered email address to identify their account, (2) "Password Textbox" where the user enters their confidential password as a masked input for security purposes, (3) "Remember Me Checkbox" that allows the user to remain logged in on their current device across sessions without needing to re-enter their credentials, (4) "Login Button" that submits the entered credentials to authenticate and grant the user access to their respective web dashboard based on their assigned role, and (5) "Forgot Password Link" that redirects the user to a password recovery page where they can reset their credentials in case they are forgotten.

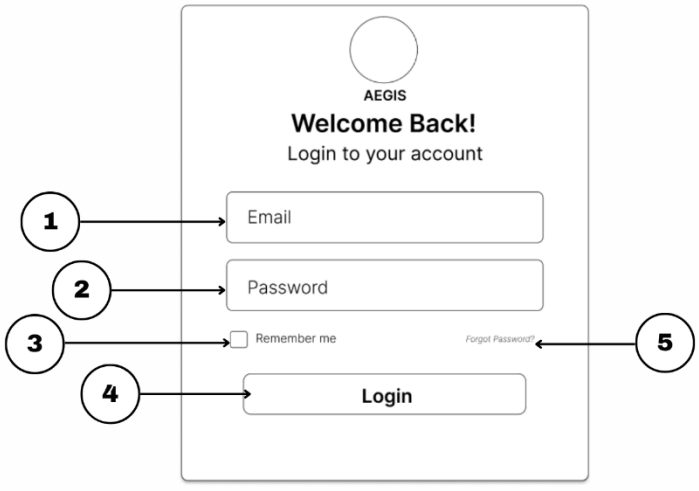


Figure [21]. Web Login Interface

1. **Email Textbox**
2. **Password Textbox**
3. **Remember Me Checkbox**
4. **Login Button**
5. **Forgot Password Link**

Figure [22] shows the Adviser/Faculty Web Dashboard interface that consists of (1) "Dashboard Navigation Button" that serves as the active home link returning the user to the main dashboard overview, (2) "Student Profiles Navigation Button" under the Advisory Class section that directs the adviser to the list of student profiles under their advisory class, (3) "Attendance Navigation Button" and "Behavioral Notes Navigation Button" under the Advisory Class section that navigate the adviser to the attendance monitoring page and the behavioral notes recording page respectively, (4) "Subject & Scheduling Navigation Button" under the Academic Records section that directs the adviser to their assigned subjects and class schedules, (5) "Grades Encoding Navigation Button" under the Academic Records section that takes the adviser to the grade encoding page where they can input and update student grades, (6) "Logout Button" that securely signs the adviser out of the system and returns them to the Login page, (7) "Total Students Card" that displays the total number of students under the adviser's advisory class, (8) "Attendance Card" that shows the current attendance count out of the total number of students for the day, (9) "Pending Grades Card" that displays the number of student grades that have not yet been encoded or finalized, (10) "Behavior Notes

Card" that shows the total number of behavioral notes recorded for the advisory class, (11) "Add Student Button" under the Quick Actions section that allows the adviser to enroll or register a new student into the system, (12) "Take Attendance Button" under the Quick Actions section that navigates the adviser to the attendance taking interface for their class, (13) "Encode Grades Button" under the Quick Actions section that redirects the adviser directly to the grade encoding page, (14) "Add Behavior Note Button" under the Quick Actions section that allows the adviser to quickly record a new behavioral observation for a student, and the dashboard also displays an "Upcoming Schedule Section" that lists the adviser's forthcoming class schedules with their corresponding time and subject, and a "Recent Notes Section" that shows the latest behavioral notes recorded by the adviser including the date, student name, and nature of the observation.

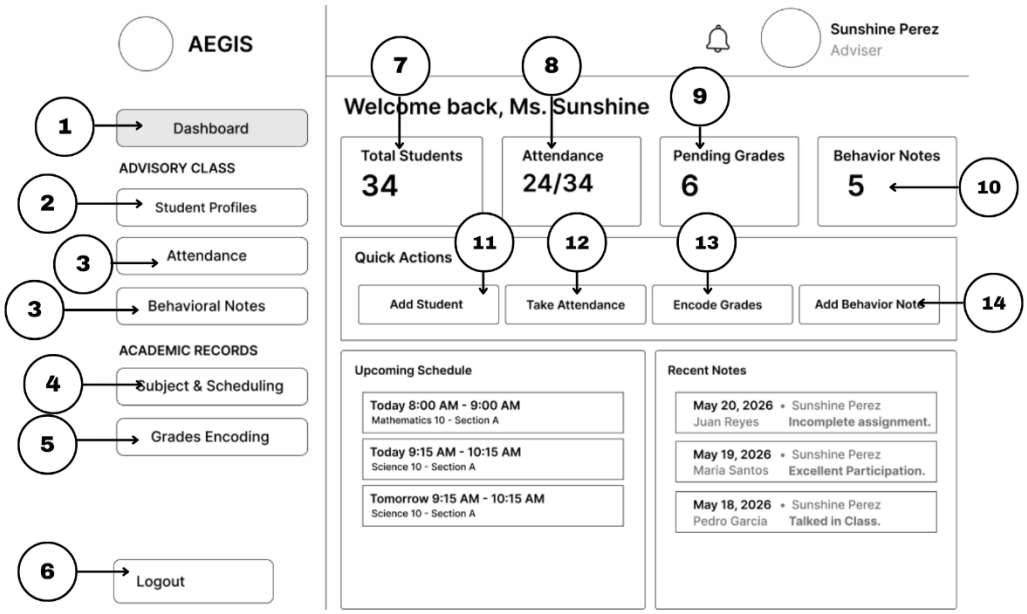


Figure [22] . Adviser/Faculty Web Dashboard

1. **Dashboard Navigation Button**
2. **Student Profiles Navigation Button**
3. **Attendance Navigation Button**
4. **Behavioral Notes Navigation Button**
5. **Subject & Scheduling Navigation Button**
6. **Grades Encoding Navigation Button**
7. **Logout Button**
8. **Total Students Card**
9. **Attendance Card**
10. **Pending Grades Card**
11. **Behavior Notes Card**
12. **Add Student Button**
13. **Take Attendance Button**
14. **Encode Grades Button**
15. **Add Behavior Note Button**
16. **Upcoming Schedule Section**
17. **Recent Notes Section**

Figure [23] shows the Student Profiles interface under the Advisory Class section that consists of (1) "Search Bar" that allows the adviser to quickly locate a specific student by typing their name or student ID, (2) "Filter Button" that enables the adviser to narrow down the student list based on specific criteria such as grade level or section, (3) "Add Students Button" that allows the adviser to register and add a new student entry into the advisory class roster, and (4) "Action Buttons"

present on each student row in the table that provide three controls — a "View Button" represented by an eye icon that opens the full profile of the selected student, an "Edit Button" represented by a pencil icon that allows the adviser to modify the student's information, and a "Delete Button" represented by a trash icon that removes the student record from the system. The table also displays each student's Student ID, Name, Grade, and Section, and Contact Number as the primary information columns for easy reference and management.

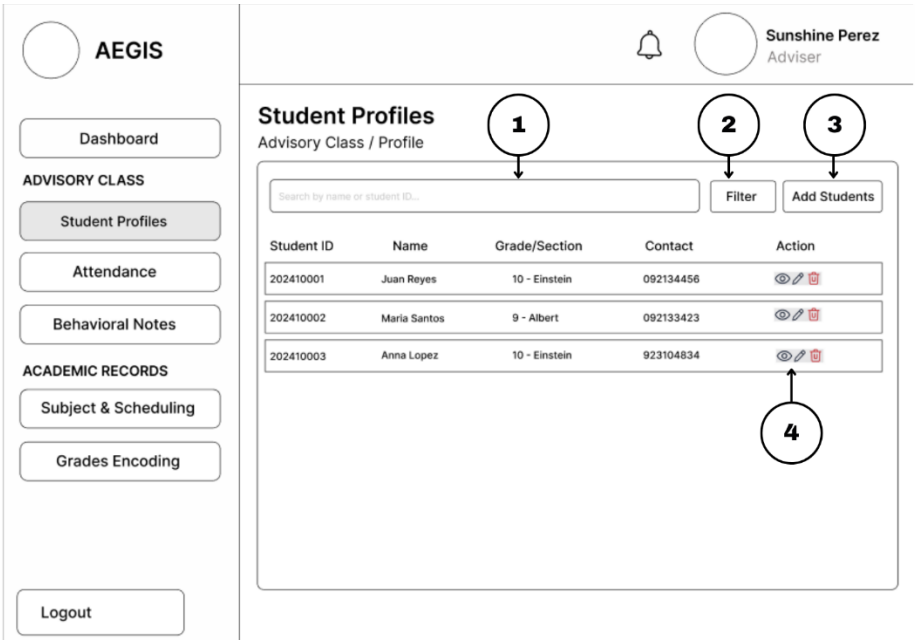


Figure [23]. Student Profiles Interface

1. **Search Bar**
2. **Filter Button**
3. **Add Students Button**
3. **View Button**
4. **Edit Button**
5. **Delete Button**

Figure [24] shows the Attendance Monitoring interface under the Advisory Class section that consists of (1) "Select Date Field" that allows the adviser to choose a specific date for which the attendance will be recorded or reviewed, (2) "Section Dropdown" that allows the adviser to select the specific class section whose attendance is being managed, (3) "Save Attendance Button" that submits and stores the recorded attendance entries for the selected date and section into the system, and (4) "Attendance Status Indicators" present on each student row in the table that allow the adviser to mark each student's attendance status using color-coded radio buttons — a green indicator for Present, an orange indicator for Late, and a red indicator for Absent. The table also displays each student's row number, Student ID, and Student Name as reference columns to facilitate accurate and efficient attendance recording.

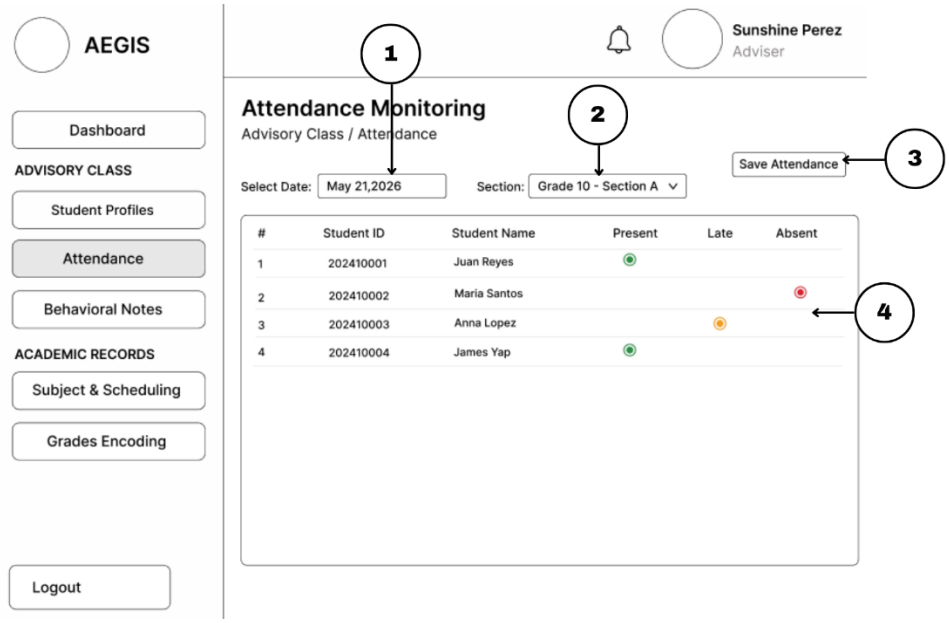


Figure [24]. Attendance Monitoring Interface

1. **Select Date Field**
2. **Section Dropdown**
3. **Save Attendance Button**
4. **Attendance Status Indicators**

Figure [25] shows the Behavior Notes interface under the Advisory Class section that consists of (1) "Select Student Field" that allows the adviser to choose the specific student for whom the behavioral note will be recorded, (2) "Student ID Field" that automatically displays the corresponding student ID upon selecting a student name, (3) "Note Type Selector" that allows the adviser to classify the behavioral note by selecting among three color-coded categories — a green "Positive" tag for commendable behavior, an orange "Warning" tag for minor concerns, and a red "Incident" tag for serious behavioral issues, (4) "Date Column"

in the notes table that displays the date on which each behavioral note was recorded, (5) "Note Column" in the notes table that shows the description or details of the recorded behavioral observation for each entry, (6) "Add New Note Textbox" that provides a text area where the adviser can type the details of the new behavioral observation they wish to record, (7) "Add Note Button" that submits and saves the newly written behavioral note along with its selected type into the student's behavioral record.

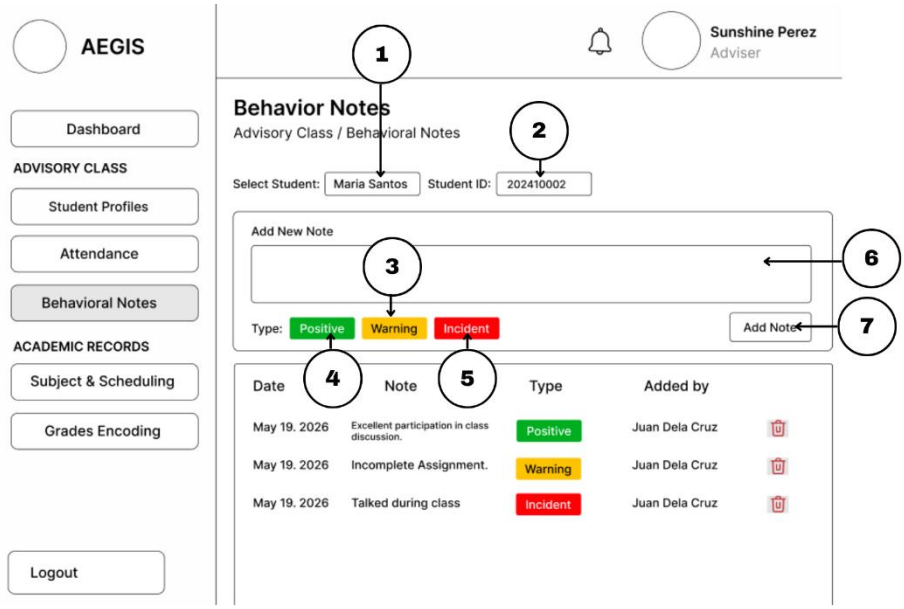


Figure [25]. Behavior Notes Interface

1. **Select Student Field**
2. **Student ID Field**
3. **Note Type Selector**
5. **Note Column**
6. **Add New Note Textbox**
7. **Add Note Button**

4. Date Column

Figure [26] shows the Subjects & Scheduling interface under the Academic Records section that consists of (1) "Action Buttons" present on each subject row in the table that provide two controls — an "Edit Button" represented by a pencil icon that allows the adviser to modify the subject details, and a "Delete Button" represented by a trash icon that removes the subject entry from the system. The table displays each subject's name, assigned teacher, schedule, and room number as the primary reference columns, listing all subjects including Mathematics 10, Science 10, English 10, Filipino 10, and Araling Panlipunan 10 with their corresponding schedules and room assignments, and (2) "Weekly Schedule Calendar" that presents a visual timetable layout organized by time slots and days of the week from Monday to Friday, displaying each subject as a color-coded block positioned according to its assigned day and time period to give the adviser a clear overview of the weekly class schedule at a glance.

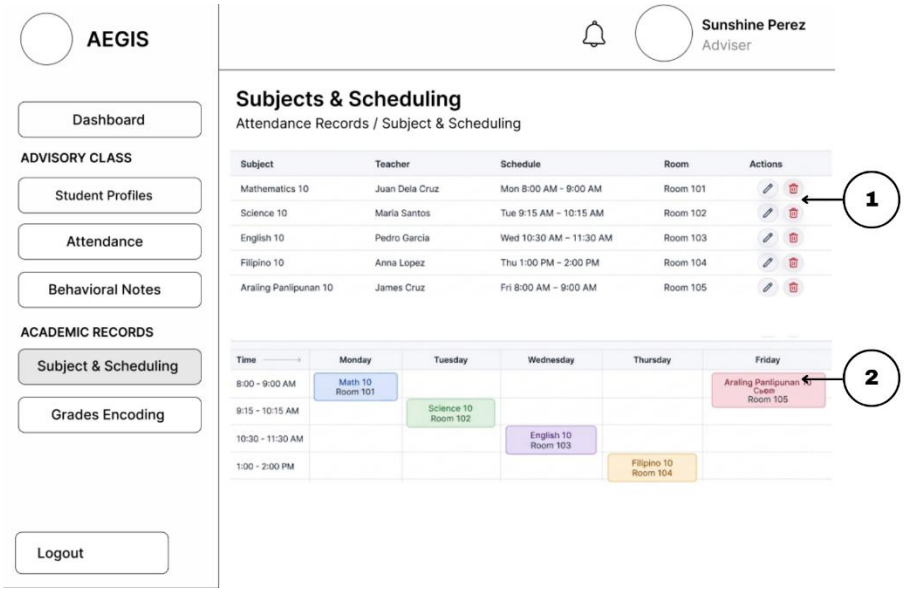


Figure [26]. Subjects & Scheduling Interface

1. **Edit Button**
2. **Delete Button**
3. **Weekly Schedule Calendar**

Figure Figure [27] shows the Grades Encoding page of the Adviser/Faculty Web Dashboard interface that consists of (1) "Subject Dropdown" that allows the adviser to select the specific subject for which grades will be encoded, currently displaying Mathematics 10, (2) "Quarter Dropdown" that enables the adviser to choose the grading period, currently set to 1st Quarter, (3) "Import Data Button" that allows the adviser to import grade data from an external file or source into the system, (4) "Save Changes Button" that saves any modifications made to the grade entries without finalizing the submission, (5) "Submit Grades Button" that finalizes and officially submits the encoded grades to the system.

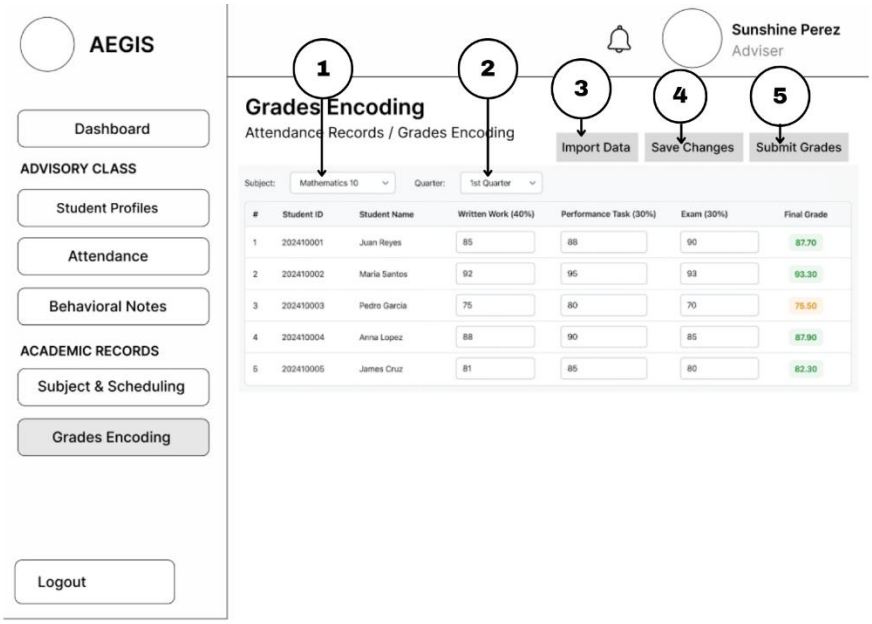


Figure 27. Grades Encoding Interface

1. **Subject Dropdown**
2. **Quarter Dropdown**
3. **Import Data Button**
4. **Save Changes Button**
5. **Submit Grades Button**

Figure [28] shows the Counselor Web Dashboard interface that consists of (1) "Dashboard Navigation Button" that serves as the active home link returning the counselor to the main dashboard overview, (2) "Student Profiles Navigation Button" that directs the counselor to the list of all student profiles in the system, (3) "Consultations Navigation Button" that navigates the counselor to the consultation records and session management page, (4) "Assessment Navigation Button" that directs the counselor to the student assessment tools and results page, (5) "AI Risk Alerts Navigation Button" that takes the counselor to the full list of AI-generated

risk alerts flagged for students requiring attention, (6) "Intervention Navigation Button" that directs the counselor to the intervention plans and tracking page for at-risk students, (7) "Settings Navigation Button" that allows the counselor to manage system preferences and account configurations, (8) "Logout Button" that securely signs the counselor out of the system and returns them to the Login page, (9) "Active Cases Card" that displays the total number of students currently under active counseling cases, (10) "At-Risk Students Card" that shows the number of students flagged by the system as at-risk requiring counselor attention, (11) "Pending Follow-up Card" that indicates the number of counseling sessions that are scheduled or awaiting follow-up action, (12) "Assessment Completion Card" that reflects the percentage and ratio of students who have completed their assessments out of the total student population, (13) "Recent Consultations Section" that presents a summarized table of the latest consultation entries showing the student name, concern, date, and current status such as Ongoing or Follow-up, accompanied by a View All link to access the complete consultation history, (14) "Upcoming Follow-up Section" that lists the counselor's scheduled sessions and conferences with their corresponding dates, times, and student names, accompanied by a View Calendar link for the full schedule, (15) "Upcoming Follow-up Entry" that represents an individual scheduled session or conference such as a parent conference or counseling session displayed as a clickable card with the date, time, and student

name, (16) "AI Risk Alert Card for High Risk Students" that highlights students flagged with a high risk score displaying the student's name, grade and section, detected risk factors such as grade decline, frequent absences, and anxiety-related chatbot queries, and recommended actions such as immediate counseling and parent coordination, along with a View Profile button, and (17) "AI Risk Alert Card for Moderate Risk Students" that similarly displays students with a moderate risk score including their detected factors such as low engagement and behavior incidents alongside recommended actions such as monitor and follow-up and encourage participation, with a corresponding View Profile button, and the interface also includes a persistent top navigation bar displaying the counselor's profile name, designation, avatar, and a notification bell icon for system alerts.

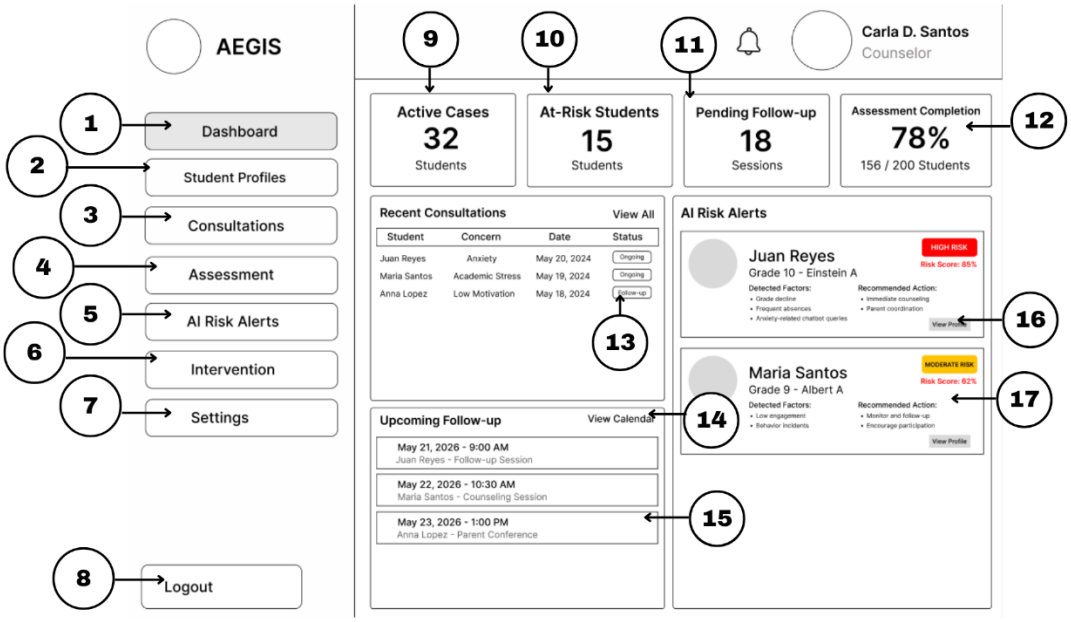


Figure [28]. Guidance Counselor Web Dashboard

1. **Dashboard Navigation Button**
2. **Student Profiles Navigation Button**
3. **Consultations Navigation Button**
4. **Assessment Navigation Button**
5. **AI Risk Alerts Navigation Button**
6. **Intervention Navigation Button**
7. **Settings Navigation Button**
8. **Logout Button**
9. **Active Cases Card**
10. **At-Risk Students Card**
11. **Pending Follow-up Card**
12. **Assessment Completion Card**
13. **Recent Consultations Section**
14. **Upcoming Follow-up Section**
15. **Upcoming Follow-up Entry**
16. **AI Risk Alert Card for High Risk Students**
17. **AI Risk Alert Card for Moderate Risk Students**

Figure [29] shows the Student Profiles page of the Counselor Web Dashboard interface that consists of (1) "Search Bar" that allows the counselor to look up a specific student by entering their name or student ID for quick access to their profile, (2) "Filter Button" that enables the counselor to narrow down the

student list based on specific criteria such as risk level, grade, or assigned counselor, (3) "Add Students Button" that allows the counselor to register or enroll a new student into the system.

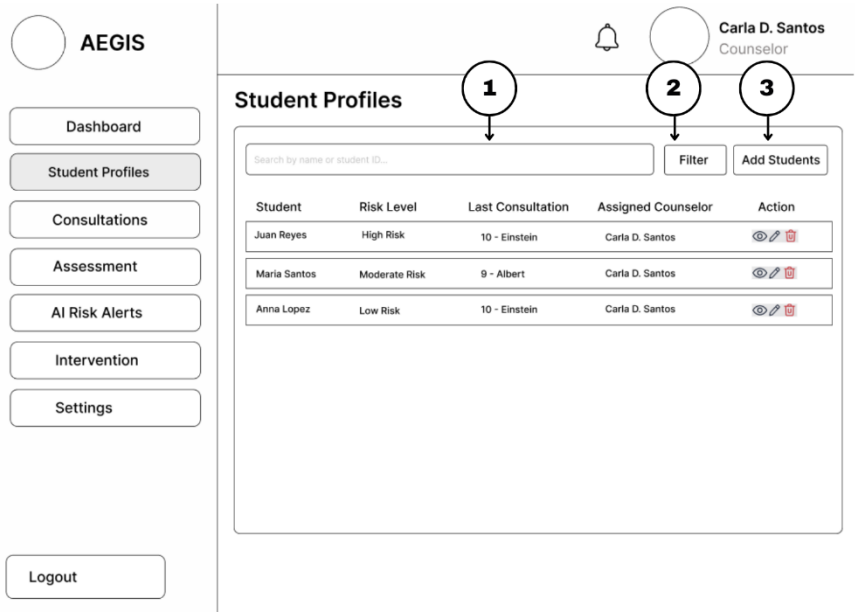


Figure [29]. Guidance Counselor Student Profiles Page

1. **Search Bar**
2. **Filter Button**
3. **Add Students Button**

Figure [30] shows the Consultations page of the Counselor Web Dashboard interface that consists of (1) "Student Information Section" that displays the selected student's profile photo, full name, student ID, grade level and section, and assigned adviser, providing the counselor with a quick overview of the student prior to

recording the consultation, (2) "View Profile Button" that redirects the counselor to the full student profile page for a more detailed view of the student's records and history, (3) "Date Field" under the Session Details section that allows the counselor to enter or select the date on which the consultation session is taking place, (4) "Time Dropdown" that enables the counselor to specify the time of the consultation session, (5) "Session Type Dropdown" that allows the counselor to categorize the nature of the session such as Individual Counseling or other session types available in the system, (6) "Counselor Dropdown" that lets the counselor assign or indicate the specific counselor handling the session, (7) "Concern Categories Checkboxes" that allow the counselor to select one or more concern categories that best describe the student's presenting issue, with options including Academic, Behavioral, Emotional, Family-related, Bullying, and Other, (8) "Next Session Date Field" under the Follow-up section that allows the counselor to schedule the date of the next follow-up session with the student, (9) "Priority Level Dropdown" under the Follow-up section that enables the counselor to set the urgency of the follow-up session with options such as High, Medium, or Low, (10) "Session Notes Text Area" that provides a free-text field where the counselor can record observations, discussions, and important details from the consultation session, (11) "Recommendations / Action Plan Text Area" that allows the counselor to document suggested interventions, action steps, or referrals based on the outcome of the

consultation, and (12) "Save Session Button" that finalizes and saves all entered consultation details into the system, accompanied by a (13) "Cancel Button" that discards any unsaved changes and exits the form.

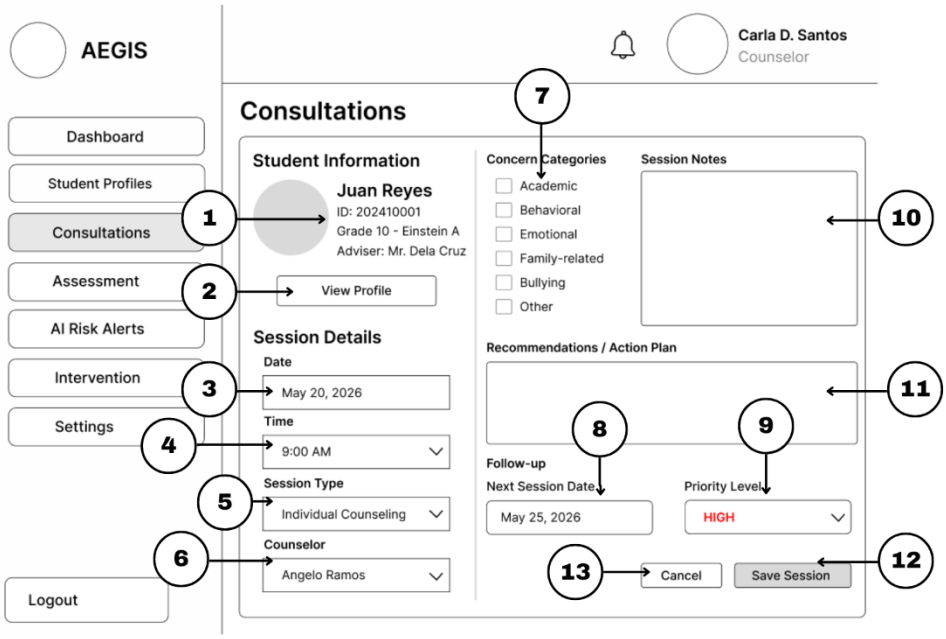


Figure [30]. Consultations Interface

1. **Student Information Section**
2. **View Profile Button**
3. **Date Field**
4. **Time Dropdown**
5. **Session Type Dropdown**
6. **Counselor Dropdown**
7. **Concern Categories Checkboxes**
8. **Next Session Date Field**
9. **Priority Level Dropdown**
10. **Session Notes Text Area**
11. **Recommendations / Action Plan Text Area**

12. Save Session Button **13. Cancel Button**

Figure [31] shows the Assessment page of the Counselor Web Dashboard interface that consists of (1) "Assessment Title Field" that displays the name of the current assessment being administered or created, shown here as Emotional Wellness Assessment, allowing the counselor to identify or modify the title of the assessment form, (2) "Add Question Button" and "Import Question Button" located at the bottom of the question area, where the Add Question button allows the counselor to manually create and append a new question to the assessment, and the Import Question button enables the counselor to bring in pre-existing questions from another source or question bank into the current assessment, (3) "AI Assistant Button" that provides the counselor with AI-powered support such as suggesting relevant questions, analyzing responses, or offering guidance in building a more comprehensive and appropriate assessment for the student, (4) "Question Display Area" that presents the current assessment question along with its corresponding answer choices in a Likert-scale radio button format ranging from 1 which is Never to 5 which is Always, allowing the counselor to review how the question will appear to the student, with a progress indicator at the top showing the current question number out of the total such as Question 3 of 12, and (5) "Next Button" and (6) "Back Button" that allow the counselor to navigate between questions within the

assessment, where the Back button returns to the previous question and the Next button proceeds to the following one.

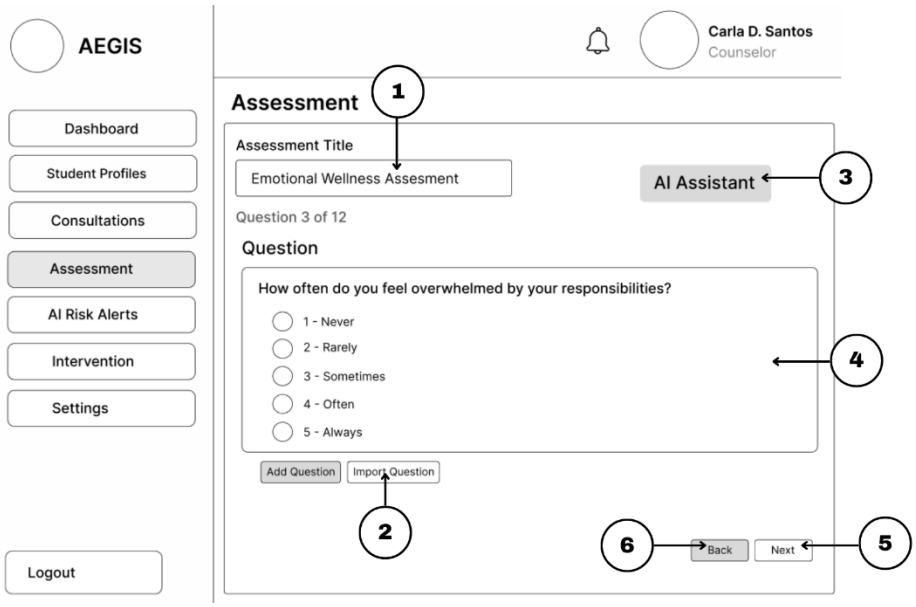


Figure [31]. Assessment Page Interface

1. **Assessment Title Field**
2. **Add Question Button**
3. **Import Question Button**
4. **AI Assistant Button**
5. **Question Display Area**
6. **Next Button**
7. **Back Button**

Figure [32] shows the AI Risk Alerts page of the Counselor Web Dashboard interface that consists of (1) "Risk Alert List Panel" on the left side that displays a scrollable list of flagged students with each entry showing the student's profile photo, full name, grade level and section, a color-coded risk level badge such as High Risk in red or Moderate Risk in yellow, a numerical risk score percentage, the

time or date the alert was generated, and a brief description of the detected risk indicators such as frequent absences and grade decline, behavioral incidents and peer conflicts, or anxiety-related chatbot queries, and upon selecting a student from the list the right panel updates to show that student's detailed alert information, (2) "Student Risk Detail Panel" on the right side that presents the selected student's profile photo, full name, grade level and section, risk level badge, a visual risk score progress bar showing the computed percentage, a Status indicator displaying the current monitoring state such as Under Monitoring, the name and role of the Assigned Counselor responsible for the student, and the date and time the alert record was last updated, and (3) "AI Recommendation Section" located at the bottom of the Student Risk Detail Panel that provides AI-generated intervention suggestions based on the student's detected risk factors, displaying a summary recommendation statement such as immediate counseling intervention and parent coordination recommended.

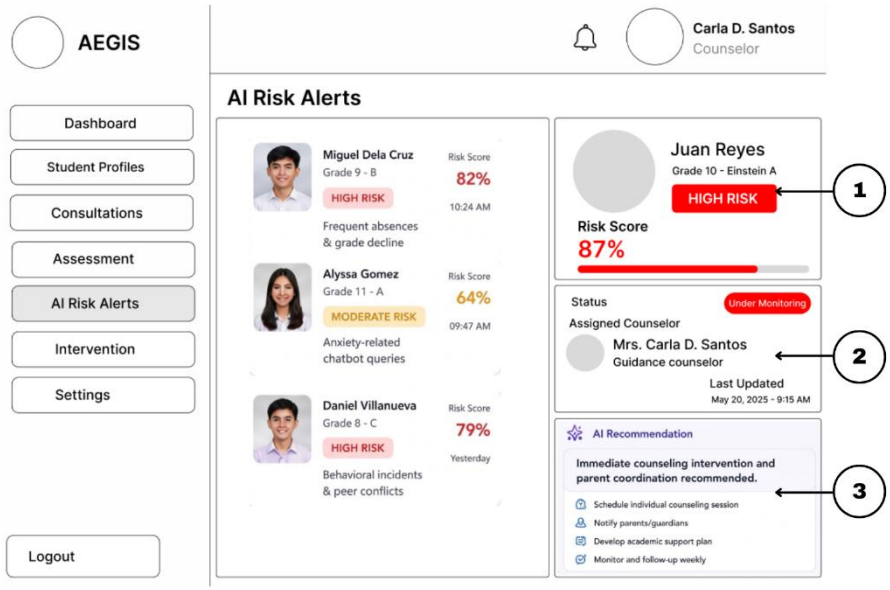


Figure [32]. AI Risk Alerts Page Interface

1. **Risk Alert List Panel**
2. **Student Risk Detail Panel**
3. **AI Recommendation Section**

Figure [33] shows the Intervention page of the Counselor Web Dashboard interface that consists of (1) "Plan Start Date Card" that displays the date on which the intervention plan was initiated, (2) "Target Review Date Card" that shows the scheduled date by which the intervention plan is expected to be reviewed or evaluated for progress, (3) "Assigned Counselor Card" that indicates the name of the counselor responsible for overseeing and implementing the intervention plan, (4) "Intervention Steps Section" that presents a numbered list of structured action items included in the intervention plan, with each step showing the step title, a brief description of the activity, and a color-coded status badge indicating the current state

of each step such as In Progress in blue for Individual Counseling which involves weekly counseling sessions, Scheduled in yellow for Parent Coordination which covers parent metings and updates, and Pending in gray for Academic Support which involves coordinating with teachers and for Monitoring and Follow-up which focuses on evaluating student progress, (5) "Progress Note Field" that provides a text area where the counselor can record observations and updates regarding the student's response to the intervention, currently showing an entry noting that the initial session was completed and that the student is open to counseling, (6) "Next Follow-up Section" that displays the date and time of the upcoming follow-up session scheduled for the student, and (7) "Save Progress Note Button" that saves the counselor's current progress note entry into the system, accompanied by a "Complete Plan Button" that marks the entire intervention plan as finished and closes the active case.

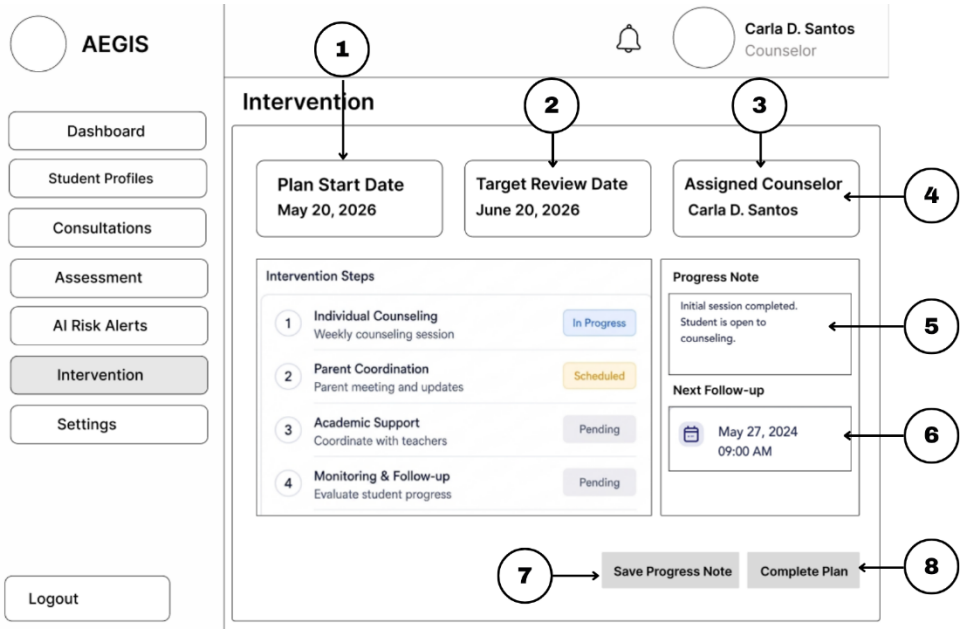


Figure [33]. Intervention Page Interface

1. **Plan Start Date Card**
2. **Target Review Date Card**
3. **Assigned Counselor Card**
4. **Intervention Steps Section**
5. **Progress Note Field**
6. **Next Follow-up Section**
7. **Save Progress Note Button**
8. **Complete Plan Button**

Figure [34] shows the Account Settings page of the Counselor Web Dashboard interface that consists of (1) "Top Navigation Bar Profile Display" that shows the counselor's full name and designation in the upper right corner of the screen, serving as a persistent identifier of the currently logged-in user across all pages, (2) "Edit Profile Button" located beneath the counselor's profile photo and

name within the Personal Information section, which allows the counselor to enter edit mode and make changes to their account details such as updating their profile picture or modifying their personal information, and (3) "Personal Information Fields" that display the counselor's current account details in read-only input fields including the First Name field showing the counselor's given name, the Last Name field showing the counselor's surname, and the Email Address field showing the registered email address associated with the account, all of which become editable upon clicking the Edit Profile button.

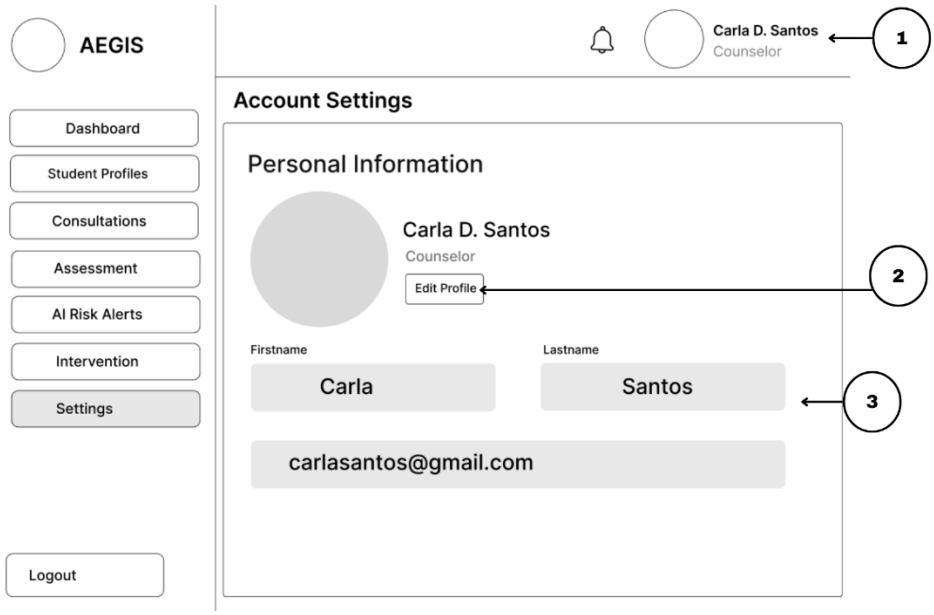


Figure 34. Account Settings Interface

1. **Top Navigation Bar Profile Display**
2. **Edit Profile Button**
3. **Personal Information Fields**

Figure [35] shows the Admin Web Dashboard interface that consists of (1) "Total Students Card" that displays the total number of enrolled students in the system with a View all Students link for accessing the complete student list, (2) "Active User Card" that shows the total number of currently active users across all roles in the system with a View all users link, (3) "Pending Imports Card" that indicates the number of data imports that are awaiting processing or review with a View imports link for quick access, (4) "Announcements Card" that displays the total number of announcements currently in the system with a View all link to access the full announcements list, (5) "System Overview Chart" that presents a donut chart visualizing the distribution of all system users broken down by role category including Students at 40%, Parents at 33%, Faculty at 12%, Guidance Counselors at 3%, and Administrators at 2%, providing the admin with a quick visual summary of the user composition across the platform, (6) "Recent Announcements Section" that displays a table of the latest announcements posted in the system.

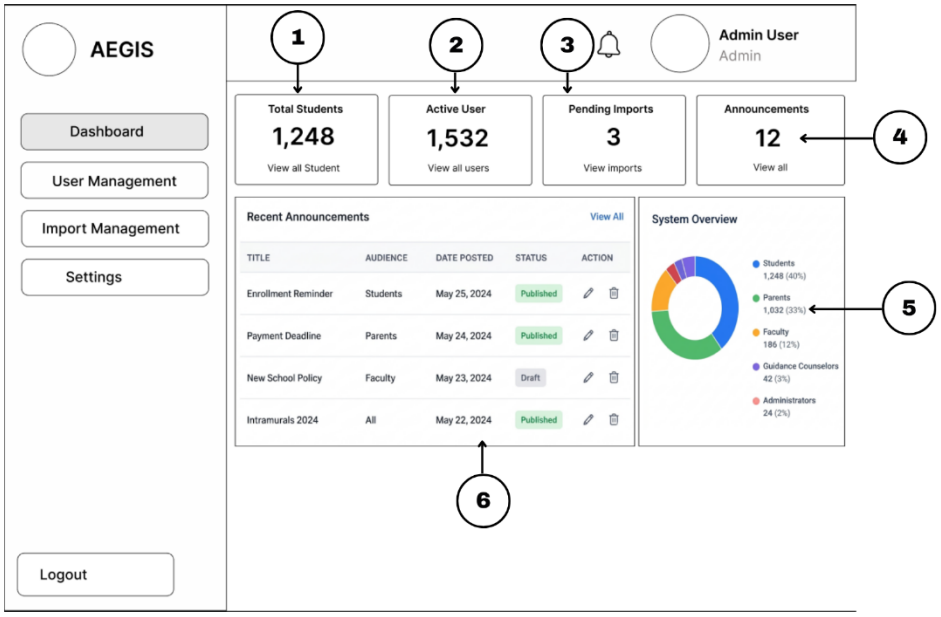


Figure 35. Admin Web Dashboard Interface

1. **Total Students Card**
2. **Active User Card**
3. **Pending Imports Card**
4. **Announcements Card**
5. **System Overview Chart**
6. **Recent Announcements Section**

Figure [36] shows the User Management page of the Admin Web Dashboard interface that consists of (1) "Search Field" that allows the admin to look up a specific user by entering their name or email address for quick retrieval from the user list, (2) "Role Dropdown" that enables the admin to filter the user list by a specific role category such as Student, Parent, Faculty, Guidance Counselor, or Administrator, currently set to display All Roles, (3) "Status Dropdown" that allows

the admin to filter users based on their current account status such as Active or Inactive, currently set to display All Status, (4) "Filter Button" that applies the selected search, role, and status criteria to narrow down and refresh the displayed user list accordingly, (5) "User Management Table" that presents a comprehensive list of all registered users in the system.

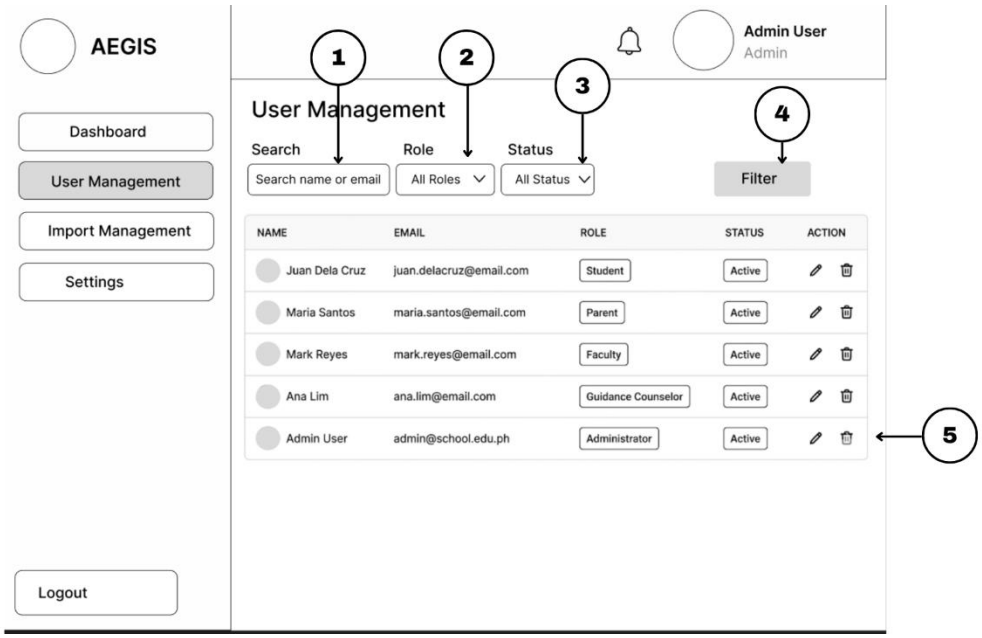


Figure [36]. User Management Page Interface

1. **Search Field**
2. **Role Dropdown**
3. **Status Dropdown**
4. **Filter Button**
5. **User Management Table**

Figure [37] shows the Data Import Management page of the Admin Web Dashboard interface that consists of (1) "File Upload Area" that provides a drag-

and-drop zone where the admin can directly drag and drop an Excel or CSV file into the designated area to begin the data import process, and (2) "Choose File Button" located within the upload area that allows the admin to manually browse and select a file from their local device for upload, with a note below indicating the supported file formats accepted by the system which include .xlsx, .xls, and .csv, etc.

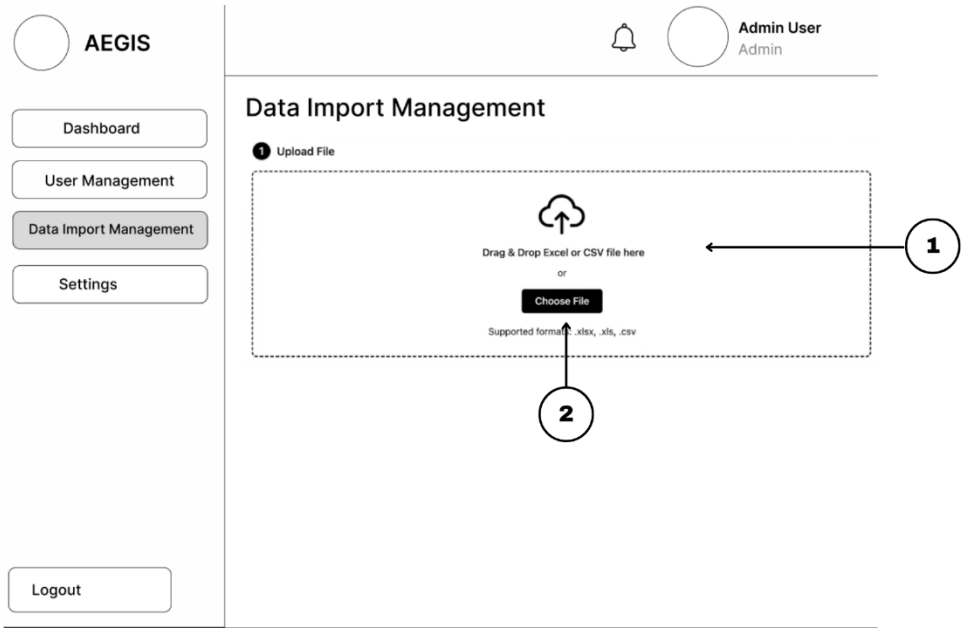


Figure [37]. Data Import Management Page Interface

1. **File Upload Area** 2. **Choose File Button**

Figure [28] shows the Settings page of the Admin Web Dashboard interface that consists of (1) "Top Navigation Bar Profile Display" that shows the admin's full name and role designation in the upper right corner of the screen, serving as a persistent identifier of

the currently logged-in administrator across all pages, (2) "Edit Profile Button" located beneath the admin's profile photo and name within the Personal Information section, which allows the administrator to enter edit mode and make changes to their account details such as updating their profile picture or modifying their personal information, and (3) "Personal Information Fields" that display the administrator's current account details in read-only input fields.

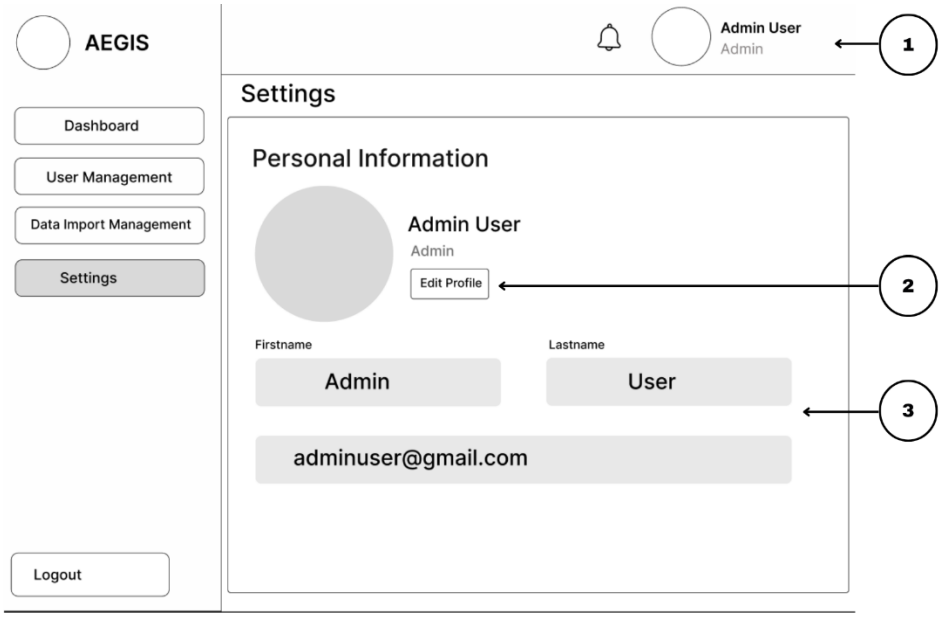


Figure – Admin Settings Page

1. **Top Navigation Bar Profile Display**
2. **Edit Profile Button**
3. **Personal Information Fields**

Hardware Requirements in Developing the System

The development of AEGIS will involve running multiple tools simultaneously, including a code editor, a local web server, a database server, and artificial intelligence components such as Natural Language Processing (NLP), Decision Tree algorithms, and chatbot integration services. The developer's machine must have sufficient hardware capacity to support these processes efficiently without experiencing significant slowdown during development, testing, and deployment. An Intel Core i5 or equivalent processor is recommended to efficiently handle compiling, debugging, database operations, and the execution of AI-related processes within the system. At least 8 GB of RAM is needed to allow multitasking and the smooth execution of development environments, browsers, machine learning libraries, and server processes. A solid-state drive (SSD) with at least 256 GB of storage is also recommended to ensure faster data access, improved loading times, and better overall system responsiveness, particularly when processing large datasets and running AI-powered features. A stable internet connection is also necessary to support the downloading of dependencies, libraries, APIs, and frameworks required throughout the development process, as well as to enable testing of the AI Chatbot, notification services, and other online components integrated within AEGIS. Additionally, a display with a resolution of at least 1080p

is advisable to ensure that the dashboards, reports, analytics, and user interfaces of the system can be previewed, designed, and tested accurately during development, especially considering the multi-module and dashboard-heavy nature of the platform.

Table 3 presents the minimum hardware specifications required for the development of AEGIS.

Table 3.

Hardware Requirements for the Developers

Hardware	Specification
Processor (CPU)	Intel Core i5 or equivalent, 2.0 GHz or higher
Memory (RAM)	8 GB or higher
Storage	256 GB SSD or higher

Hardware Requirements in Using the System

This section outlines the essential hardware components required to effectively use AEGIS. Since AEGIS is a web-based school information and management system, users can access the platform through a web browser using desktop computers, laptops, tablets, or mobile devices without the need for installing additional software. This allows the system to remain accessible and

convenient for students, parents, teachers, guidance counselors, and administrators using commonly available devices.

The hardware requirements for users are intentionally kept minimal to ensure accessibility across different types of devices and internet environments. For desktop or laptop users, the device should have at least a 1.5 GHz processor, 2 GB of RAM, and a display resolution of at least 1024 × 768 pixels to properly access dashboards, reports, announcements, and AI-powered features such as the chatbot and notification system. For mobile users, the device should have at least 2 GB of RAM and a screen size of 5 inches or larger to ensure proper viewing and interaction with the system interface. A stable internet connection is also required for both desktop and mobile users because AEGIS relies on real-time data synchronization between the frontend interface and backend database server. This ensures that academic records, attendance updates, announcements, chatbot responses, and notifications remain updated and accessible at all times. Since the system operates entirely through a web browser, no additional storage installation is required on the user's device.

Table 4 present the minimum hardware specifications required for end users to operate AEGIS.

Table 4.

Hardware Requirements For The Users

HARDWARE	SPECIFICATION (MOBILE)
Memory (ram)	2 gb or higher
Display	5 Inches Or Larger Screen Size

Software Requirements in Developing the System

This section specifies the software tools and technologies needed to develop the AEGIS system. These include the operating system, programming languages, frameworks, libraries, database system, and development tools that will be used throughout the project. Each tool was selected based on its compatibility with the system's requirements, reliability, scalability, and suitability for implementing the system's AI-enhanced features.

The development of AEGIS requires a combination of tools for frontend development, backend development, database management, artificial intelligence, chatbot integration, and system deployment. These tools must work together seamlessly to ensure smooth communication between the user interface, backend services, database layer, and AI components. Proper configuration and

compatibility of all software tools are essential in establishing a stable and functional development environment.

The frontend of AEGIS is developed using React, HTML, CSS, and JavaScript to create an interactive and responsive user interface. The backend is built using Django and Django REST Framework, which handle system logic, authentication, and API communication. PostgreSQL is used as the primary database management system for secure and efficient storage of academic and user data.

For the system's AI and analytics features, Python is used together with Scikit-learn and NLP libraries to support the Decision Tree Algorithm and data processing tasks. The chatbot component is implemented using Botpress or Rasa, integrated with the Gemini API for natural language understanding and conversational support.

Nginx is used as the web server and reverse proxy to handle client requests and improve system performance. Git and GitHub are used for version control and collaborative development.

Table 6 presents the software requirements for the development environment.

TABLE 6.

Software Requirements For The Developers

SOFTWARE	SPECIFICATION
Operating System	Windows 10 (64-bit), macOS, or Linux (Ubuntu 20.04 or higher)
Code Editor	Visual Studio Code
Frontend Development	HTML, CSS, JavaScript
Backend Development	PHP
Database	PostgreSQL
Machine Learning	Python with Scikit-learn, XGBoost, and SHAP
Chatbot	Botpress

Requirements in Using the System

This section provides the software requirements needed to properly use AEGIS. Since AEGIS is a web-based school information and management system, users do not need to install any specialized software. They only need a modern web browser and a stable internet connection.

The system is designed to work on both desktop and mobile devices. For desktop users, any modern web browser will support the platform. For mobile users, the system can be accessed through the device's built-in browser.

Table 7 presents the software requirements for end users to access and use AEGIS.

TABLE 7.

Software Requirements For The Users

Software	Specification
Web Browser (Desktop)	Google Chrome (latest version), Mozilla Firefox (latest version), or Microsoft Edge (latest version)
Web Browser (Mobile)	Chrome for Android or Safari for iOS (latest versions)
Connectivity	Stable internet connection (required for real-time features and cloud services)

Testing and Evaluation

This section presents the testing and evaluation procedures that will be used to assess the performance, functionality, and overall quality of AEGIS: AI-Enhanced Guidance and Integrated School Information System for HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas. Testing is a critical phase in the development process as it ensures that the system operates accurately, reliably, and in accordance with the defined requirements before it is deployed for actual use.

The proponents will conduct testing and evaluation of AEGIS using two methods. The first is test case evaluation, which will be used to verify that each functional requirement of the system performs as expected. Test cases will be designed to cover all major system functions including user authentication, academic record management, guidance documentation, AI-driven at-risk student detection, chatbot interaction, announcement management, and report generation. Each test case will identify the specific function being tested, the expected output, and the actual result, allowing the developers to detect and resolve any errors or inconsistencies before deployment.

The second method is the evaluation of the system using the ISO/IEC 25010 2011 software quality evaluation criteria. This internationally recognized standard will be used to measure the overall quality of AEGIS based on the following criteria: Functionality, which assesses whether the system performs all its intended features correctly and completely; Usability, which evaluates how easy and intuitive the system is for all types of users including administrators, faculty members, guidance counselors, parents, and students; Efficiency, which measures how well the system performs its functions within acceptable response times and resource usage; and Reliability, which determines the system's ability to maintain stable and consistent performance under normal operating conditions.

The results of both the test case evaluation and the ISO/IEC 25010 assessment will be used to determine the effectiveness of AEGIS and to identify areas for further improvement prior to full deployment at HIS CARE CHRISTIAN INSTITUTE.

Table 8 shows the Likert scale used to evaluate how users will rate the system using the survey questionnaire. The scale that the researchers included acts as a guide for how respondents will assess the system in the context of the specified features. This will aid in evaluating the accuracy and reliability of the proposed approach.

TABLE 8.
Likert Scale

Scale	Specification
5	Strongly Agree
4	Agree
3	Fair
2	Disagree
1	Stongly Disagree

Table 9 shows a range of matching verbal interpretations. The respondents' comments in the survey questionnaire after using the program will help the

researchers determine whether the results are consistent with the right verbal interpretation.

A weighted mean of 4.20 to 5.00 is considered as Strongly Agree. Meanwhile, a weighted mean of 3.40 to 4.19 indicates Agree. A score of 2.60 to 3.39 is classified as neutral. A weighted mean of 1.80 to 2.59 is regarded as Disagree, whilst a value of 1.00 to 1.79 is classified as Strongly Disagree. These verbal interpretations serve as a standardized guide that allows the researchers to consistently and objectively evaluate the survey results, ensuring that the findings accurately reflect the level of user satisfaction and system quality across all measured dimensions.

TABLE 9.

Scale Range and Verbal Interpretation

Mean Range	Verbal Interpretation
4.20 – 5.00	Strongly Agree
3.40 – 4.19	Agree
2.60 – 3.39	Fair
1.80 – 2.59	Disagree
1.00 – 1.79	Stongly Disagree

Test Case

This section presents the test cases used to verify that each functional requirement of AEGIS performs accurately and as intended. A test case is a set of conditions or inputs used to determine whether a specific system function operates correctly and produces the expected output. Each test case is designed to cover a specific feature or process within the system, and the results are documented to identify any errors or inconsistencies that need to be resolved before deployment.

Table 9 presents the sample test case results for the Admin, which cover functions such as user authentication, importing school files, managing user accounts, system configuration, and announcement deployment. Table 10 presents the sample test case results for the Faculty, which include functions such as faculty login, uploading and importing class records, managing class records, generating reports, and receiving alerts and announcements. Table 11 presents the sample test case results for the Guidance Counselor, which cover functions such as guidance login, managing consultation records, receiving AI-generated alerts for at-risk students, creating student assessments with AI, deploying student assessments, generating student assessment reports, and exporting student assessment report files. Table 12 presents the sample test case results for the Parent, which include functions such as parent login, viewing and monitoring the child's performance and behavior, and generating performance reports. Table 13 presents the sample test

case results for the Student, which cover functions such as student login, monitoring academic records, taking assessments, receiving alerts and announcements, and interacting with the AI chatbot.

Table 10.

Sample Test Case Result for Admin

Task ID	TASK DESCRIPTION	EXPECTED RESULTS	ACTUAL RESULTS	Status
1	Admin enters valid username and password	Admin is successfully logged in and redirected to the Admin Dashboard		
2	Admin enters invalid username or password.	System displays an error message and denies access		
3	Admin uploads academic records file	Academic data is successfully imported and stored in the system		
4	Admin uploads attendance records file	Attendance data is successfully imported and stored in the system		
5	Admin uploads guidance record file	Guidance record data is successfully imported and stored in the system		
6	Admin creates a new faculty account	Faculty account is successfully created and stored in the system		
7	Admin creates a new student account	Student account is successfully created and stored in the system		
8	Admin creates a new parent account	Parent account is successfully created and stored in the system		

Table 10.
Sample Test Case Result for Student (cont'd)

9	Admin creates a new guidance counselor account	Guidance counselor account is successfully created and stored in the system		
10	Admin assigns role access to a user	Role-based access is successfully configured for the selected user		
11	Admin configures system settings	System settings are successfully updated and applied		
12	Admin creates a new school announcement	Announcement is successfully created and stored in the system		
13	Admin publishes a school announcement	Announcement is successfully deployed and made visible to all users		

Table 11 presents the sample test case results for the Faculty of AEGIS. The test cases cover the core functions available to faculty members within the system, including user authentication, uploading and importing class records, managing advisory class records, generating class reports, and receiving alerts and announcements. Each test case was designed to verify that the system correctly processes faculty inputs and produces the expected outputs, ensuring that all faculty-related functions operate accurately and reliably before the system is deployed for actual use at HIS CARE CHRISTIAN INSTITUTE.

Table 11.

Sample Test Case Result for Faculty

Task ID	TASK DESCRIPTION	EXPECTED RESULTS	ACTUAL RESULTS	Status
1	Faculty enters valid username and password	Faculty is successfully logged in and redirected to the Faculty Dashboard		
2	Faculty enters invalid username or password	System displays an error message and denies access		
3	Faculty uploads academic records file	Academic data is successfully imported and stored in the system		
4	Faculty uploads attendance records file	Attendance data is successfully imported and stored in the system		
5	Faculty views the list of students under their advisory class .	List of students is successfully retrieved and displayed		

Table 11.

Sample Test Case Result for Administrators (cont'd)

ISO Software Evaluation

In assessing the AEGIS system under the ISO/IEC 25010:2011 quality standard, its performance is evaluated in terms of functionality, usability, efficiency, and reliability to determine how well it meets the needs of the administrators, faculty members, and guidance counselor of HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas. This evaluation focuses on how effectively the system supports academic record management, student monitoring, guidance documentation, AI-driven at-risk student detection, AI-assisted chatbot interaction, announcement management, and report generation.

In terms of functionality, the AEGIS system is designed to provide complete and accurate features that support the daily academic and administrative operations of the school. It enables administrators to import school files, manage user accounts, configure system settings, and deploy school announcements. Faculty members can upload and manage class records, generate class reports, and receive alerts and announcements through the system. The guidance counselor can document consultation records, receive AI-generated alerts for at-risk students, create and deploy student assessments using AI, generate student assessment reports, and export assessment output files. The system also provides students and parents with

their own access portals where they can view academic records, monitor performance and behavior, and receive school updates independently.

Regarding efficiency, the system operates effectively under normal network conditions. The dashboard loads and displays relevant data for each user role without significant delay. The AI-driven academic monitoring component processes student data and generates at-risk classifications within a reasonable time frame. The AI-assisted chatbot responds to student concerns promptly and delivers counselor-approved recommendations without causing system slowdowns. The system is designed to handle multiple concurrent users, ensuring that administrators, faculty members, guidance counselors, parents, and students can access the platform simultaneously without experiencing performance degradation.

In terms of usability, AEGIS offers a simple and intuitive interface suitable for all types of users, including those with limited technical backgrounds. The design supports easy navigation, clear data entry forms, and organized role-specific dashboards. The layout is consistent across all pages, and information is presented in a structured and readable manner. The responsive design ensures that the platform works across different devices and browsers, allowing users to access the system conveniently whether they are using a desktop computer or a mobile phone.

For reliability, the system ensures consistent and stable performance by safeguarding data integrity across all operations. All user inputs are validated to prevent incorrect or incomplete data entry. The AI-driven academic monitoring component produces consistent at-risk classifications when given the same input data, and the chatbot delivers consistent responses to identical student concerns. The system is designed to operate without unexpected crashes or errors during normal use, ensuring that school operations are not disrupted by system instability.

Overall, the AEGIS system demonstrates acceptable and consistent performance across the ISO/IEC 25010:2011 quality criteria of functionality, usability, efficiency, and reliability. Although further improvements may still be considered, particularly in terms of AI model accuracy and chatbot response coverage, the system effectively supports academic record management, student monitoring, and guidance services, promoting a more organized, proactive, and data-driven approach to school operations at HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas

Method of Testing

A structured testing strategy will be applied to ensure the effectiveness and usability of the AEGIS system. This will include a combination of black-box testing, exploratory testing, and user acceptance testing.

Black-box testing will be used to evaluate the system's functions from the perspective of the users. This approach focuses on inputs and outputs without requiring knowledge of the system's internal code structure. For AEGIS, black-box testing will verify that user actions, such as logging into the system, importing school files, managing user accounts, uploading and managing class records, generating class reports, documenting consultation records, receiving AI-generated alerts for at-risk students, creating and deploying student assessments, interacting with the AI-assisted chatbot, and accessing role-specific dashboards, produce the expected results. It will also validate that administrative functions, such as configuring system settings, managing role-based access control, and deploying school announcements, work correctly.

Exploratory testing will allow testers to freely interact with the system to identify possible issues, usability concerns, and unexpected errors that may not be detected through predefined test cases. Testers will simulate real-world scenarios that administrators, faculty members, and guidance counselors might encounter during actual school operations, such as importing multiple files in succession, managing several user accounts at once, navigating between different modules, generating reports for multiple students, receiving and responding to AI-generated

at-risk alerts, and deploying announcements to all user roles. This approach will help uncover edge cases and usability problems that structured test cases might miss.

User acceptance testing will involve actual users to validate whether the system meets their needs and expectations. The administrators, faculty members, and guidance counselor of HIS CARE CHRISTIAN INSTITUTE of Santa Maria, Sto. Tomas, Batangas will be invited to use AEGIS during a trial period. After using the system, participants will answer a survey questionnaire based on the ISO/IEC 25010:2011 quality model, focusing on functionality, usability, efficiency, and reliability. The feedback gathered will be used to determine whether AEGIS effectively supports academic record management, student monitoring, guidance documentation, and school communication in real-world school conditions. Any issues or suggestions raised by the participants will be documented and addressed before the final deployment of the system.

Project Teams and their Responsibilities

The completion of this project depends on the coordinated work of the research team, where each member takes on a specific role based on their skills and strengths. Defining these roles clearly helps maintain organized workflow, efficient collaboration, and accountability at every stage of development. The following

contexts will present the project members, their roles, and the tasks they will perform to contribute to the successful completion of the system.

Mr. Justine Lay J. Villegas will serve as the project manager and developer. He will lead the overall direction of the project, oversee planning and execution, and coordinate communication among team members to ensure smooth workflow and efficient collaboration.

Ms. Stephanie Loreine M. Villanueva will be responsible for UI design, frontend development, and quality assurance. She will design the user interface, implement the frontend, and conduct testing procedures to identify and resolve system issues, ensuring that the system is reliable and user-centered.

Mr. Jharen B. Narvacan will handle testing, documentation, and UI design. He will assist in designing the interface, prepare and organize all project documentation including system requirements and user manuals, and help verify that the system functions correctly and meets the needs of its target users.

Through their defined roles and collaborative efforts, the team aims to deliver a fully functional, reliable, and user-friendly system that effectively supports students living away from their families at Batangas State University – JPLPC Malvar Campus.

Schedule and Timeline

The development of AEGIS followed a structured timeline to ensure that each phase of the project was properly organized and completed in a timely manner. The process began with Brainstorming, where the team identified the problem, discussed the concept of the system, and established the overall direction of the study. This was followed by Requirements Gathering and Planning, during which the researchers collected the necessary data from respondents and outlined the objectives, scope, and features of the proposed system.

Once the groundwork was established, the team proceeded to document the study through Chapter 1, Chapter 2, and Chapter 3, covering the project context, related systems and literature, and the system design and methodology respectively. These were completed prior to the Preliminary Defense, which served as the formal presentation of the study before the panel of examiners.

After the defense, the team will move forward to the Development Phase, where AEGIS will be built and refined following the Agile methodology. This will be followed by the Testing Phase to ensure that all system features function correctly and meet the needs of the intended users. The results will then be documented in Chapter 4 and Chapter 5, and the study will conclude with the Deployment Phase,

where the completed system will be made available to its target users at Batangas

Task	Start Date	End Date	Feb	March	April	May	June	July	Aug	Sept	Oct	Nov	Dec
Brainstorming	02/20//26	03/10/26											
Requirements Gathering	03/18/26	04/23/26											
Planning	03/13/26	04/24/26											
Chapter 1	04/24/26	04/30/26											
Chapter 2	04/24/26	04/30/26											
Chapter 3	04/24/26	04/30/26											
Preliminary Defense	05/05/26	05/05/26											
Development Phase													
Testing Phase													
Chapter 4													
Chapter 5													
Deployment Phase													

Figure 34. Project Timeline of AEGIS

BIBLIOGRAPHY

[1] Mfanafuthi Dube, Thandi Cele, and Sipho Mokoena. 2025. Exploring the Impact of Information and Communication Technology on Educational Administration: A Systematic Scoping Review. Education Sciences 15, 9 (2025), 1114. https://doi.org/10.3390/educsci15091114

[2] Pedro R. Gonzalez, Maria T. Santos, and Juan A. Reyes. 2022. Parental Involvement and Academic Achievement Among Secondary School Students: A Review of Evidence. Journal of Educational Research 115, 3 (2022), 145-162. https://doi.org/10.1080/00220671.2022.2048569

[3] Scott Baker, Russell Gersten, and Dae-Sik Lee. 2021. A Synthesis of Empirical Research on Teaching Mathematics to Low-Achieving Students. The Elementary School Journal 103, 1 (2021), 51-73. https://doi.org/10.1086/499715

[4] J. M. Tugano, R. A. Dator, and M. A. Bation. 2022. Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines. arXiv preprint arXiv:2109.00198. https://arxiv.org/abs/2109.00198

[5] Fan Ouyang, Weiqi Xu, and Mutlu Cukurova. 2023. Data-Driven Artificial Intelligence in Education: A Comprehensive Review. IEEE Transactions on Learning Technologies 17 (2023), 12-31. https://doi.org/10.1109/TLT.2023.3314610

[6] M. S. Abdul Razak, N. A. Alias, N. H. Harun, N. A. Wahab, and N. F. Zulkifli. 2024. Enhancing Student Support with AI: A College Assistance Chatbot Using NLP and ANN. In *Proceedings of the 2024 Second International Conference on Networks, Multimedia and Information Technology (NMITCON)*. IEEE. https://doi.org/10.1109/nmitcon62075.2024.10699279

[7] Ismail O. Muraina, Edward A. Aiyegbusi, and Solomon O. Abam. 2022. Decision Tree Algorithm Use in Predicting Students' Academic Performance in Advanced Programming Course. *International Journal of Higher Education Pedagogies* 3, 4 (2022), 13–23. https://www.diamondopen.com/journals/index.php/ijhep/article/view/274

[8] Caldarini, G., Jaf, S., and MacGlashan, J. 2022. A Literature Survey of Recent Advances in Chatbots. *Information* 13, 1 (2022), 41. https://doi.org/10.3390/info13010041

[9] Schafer, J.B., Frankowski, D., Herlocker, J., and Sen, S. 2007. Collaborative Filtering Recommender Systems. In *The Adaptive Web*. Springer, Berlin, Heidelberg, 291–324. https://doi.org/10.1007/978-3-540-72079-9_9

[10] Britannica. 2026. Content management system. *Encyclopaedia Britannica*. Retrieved May 2026 from https://www.britannica.com/technology/content-management-system

[11] Republic of the Philippines. 2012. *Republic Act No. 10173: Data Privacy Act of 2012*. Official Gazette of the Republic of the Philippines. Retrieved May 2026 from https://www.officialgazette.gov.ph/2012/08/15/republic-act-no-10173/

[12] IBM. 2025. What is a Decision Tree? *IBM Think*. Retrieved May 2026 from https://www.ibm.com/think/topics/decision-trees

[13] J. M. Tugano, R. A. Dator, and M. A. Bation. 2022. Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines. arXiv preprint arXiv:2109.00198. https://arxiv.org/abs/2109.00198

[14] IBM. 2025. What is Natural Language Processing? *IBM Think*. Retrieved May 2026 from https://www.ibm.com/think/topics/natural-language-processing

[15] Ferraiolo, D.F., Sandhu, R., Gavrila, S., Kuhn, D.R., and Chandramouli, R. 2001. Proposed NIST Standard for Role-Based Access Control. *ACM Transactions on Information and System Security* 4, 3 (2001), 224–274. https://doi.org/10.1145/501978.501980

[16] Das, K.L., Peeraiah, G., Akash, K., Paswan, U.K., and Nayak, S. 2023. Student Information System. *International Journal of Engineering Research and Technology* 11, 8 (2023). https://www.ijert.org/student-information-system

[17] Nikolaidou, M. and Anagnostopoulos, D. 2006. A Systematic Approach for Configuring Web-Based Information Systems. *Distributed and Parallel Databases* 20 (2006), 5–28. https://doi.org/10.1007/s10619-005-6832-0

[18] Carlos González-Sancho and Stéphan Vincent-Lancrin. 2016. Transforming education by using a new generation of information systems. Policy Futures in Education 14, 6 (2016), 741–758. DOI:https://doi.org/10.1177/1478210316649287

[19] J. M. Tugano, R. A. Dator, and M. A. Bation. 2022. Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines. arXiv preprint arXiv:2109.00198. https://arxiv.org/abs/2109.00198

[20] R. Salvador, P. Ramos, and J. Villanueva. 2021. Web-Based Student Monitoring and Attendance Management System for Secondary Schools. Intrnational Journal of Advanced Computer Science and Applications 12, 5 (2021), 233–240. https://doi.org/10.14569/IJACSA.2021.0120529

[21] M. Santos and A. Rivera. 2020. Development of a Web-Based Guidance Management System for Student Counseling Services. International Journal of Scientific and Technology Research 9, 4 (2020), 1523–1528. Retrieved from http://www.ijstr.org/final-print/apr2020/Development-Of-A-Web-based-Guidance-Management-System-For-Student-Counseling-Services.pdf

[22] J. Dela Cruz, R. Mendoza, and P. Garcia. 2022. Parent-School Communication Portal for Private Educational Institutions in the Philippines. International Journal of Computer Applications 184, 42 (2022), 15–21.

https://doi.org/10.5120/ijca2022922015

[23] Chinyelugo Fidelia Agada and Thecla Amoge Eze. 2024. Information and Communication Technology as a Tool for Effective School Record Management. ESUT Journal of Education 6, 2 (2024). Retrieved from https://www.esutjoe.org/index.php/esutjoe/article/view/63

[24] Carlos González-Sancho and Stéphan Vincent-Lancrin. 2016. Transforming education by using a new generation of information systems. Policy Futures in Education 14, 6 (2016), 741–758. DOI:https://doi.org/10.1177/1478210316649287

[25] J. M. Tugano, R. A. Dator, and M. A. Bation. 2022. Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines. arXiv preprint arXiv:2109.00198. https://arxiv.org/abs/2109.00198

[26] PowerSchool. 2025. PowerSchool Student Information System. Retrieved May 2026 from https://www.powerschool.com/classroom/student-information-system/

[27] Martin Dougiamas and Peter Taylor. 2003. Moodle: Using Learning Communities to Create an Open Source Course Management System. In Proceedings of EDMEDIA 2003 Conference. Association for the Advancement of Computing in Education, 171–178.

[28] Ellucian. 2024. Ellucian Banner Student Information System. Retrieved May 2026 from https://www.ellucian.com/products/banner-student

[29] Google for Education. 2025. Google Classroom Features and Educational Tools. Retrieved May 2026 from https://edu.google.com/products/classroom/

[30] Fan Ouyang, Weiqi Xu, and Mutlu Cukurova. 2023. Data-Driven Artificial Intelligence in Education: A Comprehensive Review. IEEE Transactions on Learning Technologies 17 (2023), 12–31. https://doi.org/10.1109/TLT.2023.3314610

[31] Ismail O. Muraina, Edward A. Aiyegbusi, and Solomon O. Abam. 2022. Decision Tree Algorithm Use in Predicting Students' Academic Performance in Advanced Programming Course. International Journal of Higher Education Pedagogies 3, 4

(2022), 13–23. https://www.diamondopen.com/journals/index.php/ijhep/article/view/274

[32] G. Caldarini, S. Jaf, and J. MacGlashan. 2022. A Literature Survey of Recent Advances in Chatbots. Information 13, 1 (2022), 41. https://doi.org/10.3390/info13010041